

Surcomplex Spectral Theory

Singular Values, Infinitesimal Rank,
and Multiscale Stability

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Abstract

Finite matrices over the surcomplex numbers admit much of the familiar spectral theory, but its justification is algebraic rather than compactness based, and its interpretation changes when singular values occupy different infinitesimal scales. We develop Hermitian and normal diagonalization, positive square roots, singular value and polar decompositions, variational principles, least squares, low-rank approximation, conditioning, and spectral-subspace perturbation over a real closed field and its complexification.

The central synthesis identifies the valuations of singular values with successive differences of the minimum valuations of minors. A positive Cauchy–Binet identity then recovers both these scales and the leading singular coefficients from the characteristic coefficients of the Gram matrix. This gives an intrinsic rank filtration, an exact root-free scale algorithm, and precision certificates valid for arbitrary-rank Hahn value groups. Worked examples separate exact rank from standard-part rank, small perturbations from gap-relative perturbations, and algebraic inverses from Hahn-summable inverse series. Nonnormal pseudospectra, regularization, and a finite tensor application complete the picture.

Divisibility of the value group is then removed. A finite splitting tree, licensed by strong Hahn summability rather than by any convergence argument, produces Hermitian and normal diagonalizations and singular-value decompositions whose every exponent lies in the subgroup generated by the input exponents. Against this, positive matrix roots genuinely cost new scales, and the cost is exactly the determinantal one: for positive semidefinite A of rank r the principal q th root generates the Hahn field on $\Gamma + \sum_k \mathbb{Z}\Delta_k(A)/q$, a finite abelian totally ramified extension of degree the order of the subgroup the classes Δ_k span in $\Gamma/q\Gamma$; it is trivial exactly when every Δ_k lies in $q\Gamma$, which is also the exact criterion for a Gram factorization $A = C^*C$ over the base field. The single scalar $\text{tr}(A^{1/q})$ is always a primitive generator of that whole extension, by a positivity mechanism rather than by genericity.

Status and scope. A self-contained research exposition relative to standard surreal normal-form and Hahn-field foundations, assembled from two separately delivered manuscripts (see [Section 1](#)). Classical results are identified as such; no priority claim or solution of a named open problem is made. Older power-series spectral theorems, Hahn support lemmas and Kummer theory are acknowledged as predecessors, and the arbitrary-rank determinantal and primitive-trace package of [Sections 10](#) and [11](#) is offered as a candidate original contribution whose priority is not certified. All matrix dimensions are ordinary finite integers. General proofs are not proof-assistant verified; no Lean formalization of any result here is included or claimed. The accompanying exact symbolic checks exercise finite examples, not arbitrary Hahn supports or the general theorems.

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1 The documentation gap and the mathematical agenda

1.1 What was inspected

This article was chosen after inspecting the documentation catalogue and relevant inventories in the repository [VladimirReshetnikov/Surreal](#), at the fixed revision

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The catalogue lists fifteen packages in surreal mathematics, surcomplex analysis and geometry, and foundations and computation [1]. The detailed polynomial-algebra inventory and its opening source sections were especially relevant [2]; the foundations inventory was also inspected [3]. This is a targeted coverage audit, not a claim to have checked every line of all archived source manuscripts.

Matrix theory is not wholly absent. The polynomial report uses an algebraic Schur inequality and a differentiating compression in its treatment of Schoenberg’s inequality, and it uses signatures of Hermite forms to count real roots. Finite residue pairings and multiplication matrices also appear in the surrounding algebra. These are substantial ingredients, but they are organized around polynomial roots and residues, not around matrices as the objects of study. The inspected inventories do not offer a systematic singular-value, conditioning, or scale-resolved rank theory.

Existing strand	What it supplies	What this chapter adds
Polynomial algebra	Root geometry, Schur-type arguments, signatures, Newton profiles	Unitary geometry, SVD, positive Gram matrices, determinantal singular scales
Analytic geometry	Finite algebras and deformation questions	Gap-sensitive spectral projections and Schur-complement effective matrices
Foundations	Set-sized Hahn workspaces; limits on transfer	Explicit finite algebraic proofs without compactness or a class-sized choice principle
Computer algebra	Representation and precision distinctions	Root-free scale elimination and certificates for resolvable rank

Table 1: The gap is an organizing theory, not the absence of every matrix lemma.

1.2 Why this is more than replacing scalars by surreals

Over the ordinary reals, a positive singular value is often tacitly regarded as a non-negligible number. Over the surreals, a matrix can have singular values at scales 1 , ω^{-2} , and $\omega^{-\omega}$. It is then simultaneously invertible, singular after taking standard parts, and indistinguishable from a lower-rank matrix at many prescribed precisions. An infinitesimal perturbation can rotate an eigenspace through an ordinary angle when its spectral gap is infinitesimal too. Conversely, an infinite entry need not imply an unstable direction if every singular value has the same scale.

The issue is therefore not just whether a decomposition exists. We need to know what its invariants measure, which conclusions survive reduction to an ordinary complex matrix, and which operations can be certified without computing an entire Hahn expansion. The key bridge is

$$\min_{|I|=|J|=k} v(\det A_{I,J}) = \sum_{j=1}^k v(\sigma_j(A)), \quad 1 \leq k \leq \text{rank } A. \quad (1.1)$$

Here the singular values are ordered by their actual positive magnitudes, whereas the valuations

are ordered in the opposite sense. This identity turns analytic-looking size information into finite determinantal algebra.

1.3 Attribution and reading route

The spectral theorem over real closed fields is classical, not a novel surreal phenomenon; a recent explicit use is Wang [7]. The variational and perturbation principles have their classical matrix-analysis counterparts, for example in Tao’s lecture notes [8], the original low-rank approximation paper of Eckart and Young [9], and the Davis–Kahan literature [11, 12]. The pseudoinverse is the classical Moore–Penrose construction [10].

The contribution here is an integrated account with proofs in the exact field setting, the determinantal and Gram-coefficient description of surreal scales, and explicit boundaries for reduction, computation, and summability. We do not claim that these consequences of existing algebra first occur here. Readers already comfortable with real-closed-field linear algebra can start at [Section 7](#); readers interested primarily in computation can proceed from there to [Sections 13](#) and [14](#).

1.4 Provenance and the second half of the article

This article is assembled from two separately delivered manuscripts, both dated 21 September 2026. Every section except [Sections 9](#) to [11](#) comes from `01-determinantal-singular-value-scales.tex`, which worked over a real closed field and, for the scale-sensitive results, in a *divisible* Hahn workspace. Those three sections come from `02-determinantal-ramification-descent.tex`, which removes that divisibility hypothesis and measures exactly what it was buying.

The second manuscript cites the first by name and declines to reclaim its content: the SVD, the determinantal singular scales of [Theorem 7.5](#) and the positive Cauchy–Binet identity of [Theorem 8.1](#) are stated here once only, in the first manuscript’s form. What the second adds is the removal of divisibility from the decompositions, and then an exact field-of-definition and degree calculation for positive matrix roots, in which the very invariant Δ_k that measures singular scales turns out to control a finite radical extension.

The two manuscripts audited the repository at different pinned revisions,

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the first and the second respectively. Both declared their audits targeted rather than line-by-line, with absence from a search result never used as proof of absence from the collection.

That second half has genuine predecessors, which it concedes rather than displaces. Adkins proved a spectral theorem for normal matrices over hermitian discrete valuation rings, including one-variable formal and convergent power series [13]; Keller and Ochsenius proved orthogonal diagonalization over iterated real Laurent-series fields and a particular infinite-rank Hahn field by recursive block splitting [14]. So neither “a spectral theorem beyond $\mathbb{R}$ ” nor “no fractional exponents in a Laurent-series eigenbasis” is claimed as new. Parusiński and Rond prove multiparameter formal spectral and singular-value results under normal-crossing discriminant hypotheses [15], and the 2026 preprint of Dai, Liang, Lu and Zhi gives a splitting-and-projector criterion over multivariable formal power-series rings [16]. A Hahn field is a different ring: a totally ordered exponent group permits monomial division that a multivariable power-series ring need not permit, and we do not claim to remove the obstructions in that latter category ([Warning 9.24](#)). The support foundations are standard Hahn–Neumann and word-order ideas [17, 18], reproved here for self-containment, and the finite extension calculation is an elementary monomial form of Kummer theory.

1.5 Which spectrum this is

The collection uses the word *spectrum* for several inequivalent things, so this article fixes its own sense once. Here $\text{spec } H$ is the algebraic *eigenvalue list* of a finite matrix over K , together with the valuations of those eigenvalues and of the singular values. It is not the differential *phase spectrum* of `docs/surcomplex/differential-equations/`, which is a multiset of purely infinite surreal parts of primitives of eigenvalues, and whose members are not numbers of the same kind as ours and are not finite angles. It is not the *prime spectrum* of a ring, the sense in which `docs/surcomplex/analytic-geometry/` discusses maximal and prime ideals. And the visible-rank levels $r_{<\theta}, r_{\leq\theta}$ of [Section 12](#), although they form a list of scales, are not a spectrum in any of these senses.

2 Fields, workspaces, and conventions

2.1 The algebraic setting

Until [Section 7](#), let F be a set-sized real closed ordered field and put $K = F(i)$, where $i^2 = -1$. We use the standard equivalent characterization that F is ordered and $F(i)$ is algebraically closed. Conjugation is $\overline{x + iy} = x - iy$, and

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in F_{\geq 0}.$$

Every positive element of F has a unique positive square root. No Archimedean property or completeness assumption is imposed.

For a matrix A , write $A^* = \bar{A}^T$. A square matrix is *Hermitian* when $A^* = A$, *normal* when $A^*A = AA^*$, and *unitary* when $A^*A = AA^* = I$. A matrix with real entries is treated inside F , with transpose in place of conjugate transpose. The integers m, n, k used as dimensions and ranks are always ordinary finite integers. Ambient row and column sizes are positive; zero-dimensional invariant subspaces and empty products are allowed. Unless explicitly stated otherwise, vector spaces and subspaces are over K .

We use ∞ only as an added valuation symbol, with $v(0) = \infty$. It is not a surreal number. Expressions such as $\infty - \infty$ will never be used.

2.2 Hahn workspaces inside the surcomplex numbers

For the scale-sensitive results, choose a set-sized additive subgroup $\Gamma \subseteq \mathbf{No}$, with its inherited order, and set

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)), \quad t^\gamma := \omega^{-\gamma}. \quad (2.1)$$

The last expression uses Conway's monomial map. It is not a convention about the ordinary real exponential. An element of K_Γ is a formal sum

$$a = \sum_{\gamma \in S} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{C},$$

where $S \subseteq \Gamma$ is a well-ordered set in increasing exponent order. Zero coefficients may be omitted. The field operations are Hahn addition and convolution. On F_Γ , the sign of a nonzero series is the sign of its coefficient at the least exponent.

No divisibility, finite-rank, countability, or Archimedean assumption is imposed on Γ .

We use two foundational facts. Conway normal forms identify these series with surcomplex numbers, and the Hahn field with divisible value group and algebraically closed coefficient field of characteristic zero is algebraically closed. Gonshor's Chapter 5 gives the normal-form and

support foundations [5]; Poonen’s Corollary 4 gives the relevant algebraic-closedness statement for Mal’cev–Neumann fields [6]. Thus *when Γ is divisible*, $K_\Gamma = F_\Gamma(i)$ is algebraically closed and F_Γ is real closed, so every result proved over a real closed F applies verbatim inside the workspace. These standard foundational theorems are inputs, not reproved here.

Remark 2.1 (What still needs divisibility). For a general Γ the field F_Γ is *not* real closed: if $\gamma \notin 2\Gamma$ then the positive element t^γ has no square root in F_Γ , because the valuation of a square lies in 2Γ . Divisibility is nevertheless needed only where a root of an *arbitrary* positive element, or an algebraic closure, is actually used: the Schur triangularization of a non-normal matrix in [Proposition 4.2](#), the positive semidefinite square root of an arbitrary positive semidefinite matrix and the Cholesky factorization in [Proposition 4.3](#) with the generalized Hermitian eigenproblem built on it, and consequently the functional calculus and pseudospectral statements that invoke an eigenvalue of an arbitrary square matrix — all of which live in the divisible hull $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, at a cost computed exactly in [Theorem 10.4](#). Everything else in this article, including the Hermitian and normal spectral theorems, the SVD, the polar decomposition, every norm and variational principle, the determinantal scales of [Theorem 7.5](#), the positive Cauchy–Binet identity of [Theorem 8.1](#), the rank filtration, the scale algorithm, the perturbation certificates, and the matrix series of [Theorem 18.1](#), holds over an arbitrary set-sized ordered abelian Γ , by the support-certified constructions of [Section 9](#).

Proposition 2.2 (Finite-data localization). *Every finite collection of surcomplex matrices has all its entries in one workspace (2.1). All the decompositions established below can be performed in that workspace.*

Proof. Take the union of the exponent supports in the real and imaginary normal forms of every entry. There are finitely many entries and every support is a set, so their union is a set. Its rational linear span is a divisible set-sized subgroup of $\mathbf{No}$. Reversing exponents to follow the convention $t^\gamma = \omega^{-\gamma}$ puts all entries in its Hahn field. Algebraic closedness supplies all roots needed in the spectral proofs; positive square roots and finite field operations remain in the same workspace. Every basis extension used below has finite length. $\square$

The same argument works for a set of scalar inputs, but we only need finite matrices here. No quantification over a class-sized set of all bases is required. A displayed theorem about matrices over $\mathbf{No}[i]$ means that the finite data are localized and the theorem is applied there. Its explicit algebraic witnesses are still witnesses in any larger workspace.

[Proposition 2.2](#) deliberately passes to a rational span, which is the smallest *divisible* workspace containing the data. There is a smaller and more informative choice.

Proposition 2.3 (Exact-support localization). *Let A be a finite surcomplex matrix. Take the union of the exponent supports of the real and imaginary Conway normal forms of all its entries, negate the exponents to follow the convention $t^\gamma = \omega^{-\gamma}$, and let $H \subseteq \mathbf{No}$ be the additive subgroup they generate. Then H is a set, every entry of A lies in K_H , and H is contained in every subgroup of $\mathbf{No}$ whose Hahn field contains all the entries.*

Proof. A subgroup generated by a set consists of the finite integer combinations of its generators, so it is a set. The remaining assertions are immediate from the definition of the support of a normal form. $\square$

The group H need not be divisible, and that is the point: the results of [Section 9](#) take place inside K_H itself, while [Sections 10](#) and [11](#) measure exactly how far a positive matrix root has to leave it. No proper-class Hahn sum is used anywhere in this article.

2.3 Order, valuation, and topology are separate structures

In a Hahn workspace the ordered modulus and the valuation are not the same thing. For $\rho > 0$, both t^ρ and $2t^\rho$ have valuation ρ , although the latter is outside the ordered disk $|z| \leq t^\rho$. We will use ordered inequalities for optimization and conditioning, and valuation inequalities for relative scale and support precision.

The principal proofs in this article use no infinite limiting process. A unit sphere is not assumed compact, a supremum is not assumed to exist, and no spectral integral is invoked. In [Section 18](#), where an infinite matrix series is used, its Hahn support is checked explicitly. Internal valuation convergence in a chosen Hahn field and the fine order/modulus topology inherited from the whole surreal class must not be conflated.

3 Hermitian geometry without compactness

For $x, y \in K^n$, define

$$\langle x, y \rangle = x^* y = \sum_{j=1}^n \bar{x}_j y_j, \quad \|x\| = \sqrt{x^* x}.$$

This convention is conjugate-linear in the first variable. Since F is ordered, a sum of squares is zero only if every term is zero. Consequently $x^* x > 0$ for $x \neq 0$.

Lemma 3.1 (Cauchy–Schwarz and orthogonal decomposition). *For $x, y \in K^n$,*

$$|x^* y| \leq \|x\| \|y\|.$$

If $x \neq 0$, the vector y decomposes uniquely as

$$y = x \frac{x^* y}{x^* x} + y_\perp, \quad x^* y_\perp = 0.$$

Every finite-dimensional subspace admits an orthonormal basis and has an orthogonal complement.

Proof. The asserted decomposition follows by substitution. Its squared norm is

$$y^* y = \frac{|x^* y|^2}{x^* x} + y_\perp^* y_\perp \geq \frac{|x^* y|^2}{x^* x},$$

which proves Cauchy–Schwarz; the case $x = 0$ is immediate. Gram–Schmidt now works on any finite linearly independent list: subtract the finitely many orthogonal components already constructed and divide the remaining nonzero vector by its positive norm. Extend a basis of the subspace to a basis of K^n and continue this process. The remaining orthonormal vectors span its orthogonal complement. $\square$

This proof still works when a norm is infinitesimal. Dividing by that norm may produce infinite coordinates at an intermediate step, but it is a legal field operation; the final unit vector has all coordinate moduli at most one. It is invalid to delete a nonzero vector merely because its standard part vanishes.

The triangle inequality follows from Cauchy–Schwarz:

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2.$$

Finite orthogonal projections therefore exist by an explicit basis formula, not by an infinite-dimensional projection theorem. If $u_1, \dots, u_k$ is an orthonormal basis of W , then

$$P_W = \sum_{j=1}^k u_j u_j^*, \quad P_W^2 = P_W = P_W^*, \quad K^n = W \oplus W^\perp. \quad (3.1)$$

The projector is independent of the chosen orthonormal basis because it acts as the identity on W and as zero on $W^\perp$.

4 Spectral, positive, and singular-value decompositions

4.1 The finite spectral theorem

Theorem 4.1 (Hermitian spectral theorem). *For every Hermitian $H \in K^{n \times n}$, there are a unitary U and $\lambda_1, \dots, \lambda_n \in F$ such that*

$$H = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^*.$$

For real symmetric H , U can be chosen orthogonal with entries in F .

Proof. The characteristic polynomial splits in K , so there is an eigenpair $Hx = \lambda x$ with $x \neq 0$. The scalar $x^* H x$ is fixed by conjugation, hence lies in F , and

$$\lambda = \frac{x^* H x}{x^* x} \in F.$$

Normalize x . If $y \perp x$, then $x^* H y = (Hx)^* y = \lambda x^* y = 0$, so $x^\perp$ is invariant under H . The restriction to that complement is Hermitian. Induction on n completes the construction. For a real symmetric matrix, once $\lambda \in F$ is known, singularity of $H - \lambda I$ over F supplies a nonzero kernel vector in F^n . The same induction can therefore be carried out over F . $\square$

Proposition 4.2 (Schur form and normal matrices). *Every $A \in K^{n \times n}$ is unitarily similar to an upper triangular matrix. It is unitarily diagonalizable if and only if it is normal.*

Proof. Choose and normalize an eigenvector and extend it to an orthonormal basis. The first column below the diagonal then vanishes. Apply induction to the remaining square block to obtain upper triangular Schur form T . If T is normal, comparison of the $(1, 1)$ entries of TT^* and T^*T gives

$$|t_{11}|^2 + \sum_{j>1} |t_{1j}|^2 = |t_{11}|^2.$$

Positivity forces the rest of the first row to vanish. Induction on the remaining normal block makes T diagonal. Conversely a diagonal matrix is normal, and unitary conjugation preserves normality. $\square$

For distinct eigenvalues λ of a Hermitian matrix, its spectral projectors are finite polynomials:

$$P_\lambda = \prod_{\substack{\mu \in \operatorname{spec} H \\ \mu \neq \lambda}} \frac{H - \mu I}{\lambda - \mu}, \quad H = \sum_{\lambda \in \operatorname{spec} H} \lambda P_\lambda. \quad (4.1)$$

Evaluation on the diagonal form proves the formula. The eigenspace projector is canonical; a basis within a repeated eigenspace is not.

4.2 Positivity, square roots, and congruence

A Hermitian matrix is positive semidefinite, written $H \succeq 0$, when $x^* H x \geq 0$ for every x , and positive definite, written $H \succ 0$, when this is strict for $x \neq 0$. These are equivalent respectively to all eigenvalues being nonnegative or positive. In particular, positive definite does not mean bounded below by a positive *ordinary real* multiple of I .

Proposition 4.3 (Positive square root and inertia). *A positive semidefinite matrix has a unique positive semidefinite square root. The numbers of positive, negative, and zero eigenvalues of a Hermitian matrix are unchanged by an invertible congruence $H \mapsto S^* H S$.*

Proof. Diagonalize H and replace each eigenvalue by its nonnegative square root. For uniqueness, a positive semidefinite R with $R^2 = H$ commutes with H , so it preserves each eigenspace of H . On the eigenspace with eigenvalue λ , diagonalization of R gives only nonnegative roots of $r^2 = \lambda$, hence $R = \sqrt{\lambda}I$ there.

If p is the number of positive eigenvalues of H , its positive eigenspace is a p -dimensional subspace on which the quadratic form is positive definite. Any subspace of dimension greater than p meets the span of the nonpositive eigenvectors nontrivially, so cannot have that property. Thus p is an intrinsic maximum dimension, preserved by invertible congruence. Apply the same argument to $-H$ for the negative count; subtract both counts from n for the nullity. $\square$

The usual Cholesky factorization also survives. If $H = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \succ 0$, then $a > 0$ and $C - bb^*/a \succ 0$: evaluate the form at $(-a^{-1}b^*y, y)$. Inductively factor that Schur complement as LL^* . Then

$$H = \begin{pmatrix} \sqrt{a} & 0 \\ b/\sqrt{a} & L \end{pmatrix} \begin{pmatrix} \sqrt{a} & b^*/\sqrt{a} \\ 0 & L^* \end{pmatrix}.$$

This is a finite algorithm, not a limiting process.

4.3 Singular values and the polar decomposition

Theorem 4.4 (Singular value decomposition). *Let $A \in K^{m \times n}$ and $p = \min(m, n)$. There are unitaries U, V and a rectangular diagonal matrix Σ such that*

$$A = U\Sigma V^*, \quad \sigma_1 \geq \cdots \geq \sigma_r > 0, \quad \sigma_{r+1} = \cdots = \sigma_p = 0,$$

where $r = \text{rank } A$ and the diagonal entries of Σ are the σ_j . The positive σ_j are uniquely determined with multiplicity.

Proof. The matrix A^*A is positive semidefinite. Choose an orthonormal eigenbasis $v_1, \dots, v_n$ and write its positive eigenvalues as $\sigma_1^2, \dots, \sigma_r^2$. For $j \leq r$, put $u_j = Av_j/\sigma_j$. Then

$$u_j^* u_k = (\sigma_j \sigma_k)^{-1} v_j^* A^* A v_k = \delta_{jk}.$$

When the eigenvalue is zero, $\|Av_j\|^2 = 0$ implies $Av_j = 0$. Extend the u_j to an orthonormal basis of K^m . These bases give the claimed factorization, with precisely r nonzero diagonal entries. The characteristic polynomial of A^*A determines the squared singular values, and the positive square root determines the singular values themselves. $\square$

When convenient we extend the list by zeros up to length n , so that σ_j^2 lists all eigenvalues of A^*A . In particular, formulas using input subspaces of dimension k remain meaningful for $k > p$.

Set $|A| = (A^*A)^{1/2}$. The SVD gives the polar factorization

$$A = W|A|, \quad W = \sum_{j=1}^r u_j v_j^*. \quad (4.2)$$

Here W is the unique map that is an isometry from $(\ker A)^\perp$ to $\text{im } A$, is zero on $\ker A$, and satisfies (4.2). On that complement $|A|$ is invertible, so the uniqueness follows directly. For a square invertible matrix, W is unitary. For a singular square matrix, unitary extensions of W exist but are generally not unique. Confusing the canonical partial isometry with one of those extensions loses information.

4.4 A generalized Hermitian eigenproblem

If $G \succ 0$ and $H = H^*$, the equation $Hx = \lambda Gx$ reduces to the ordinary Hermitian problem for

$$C = G^{-1/2} H G^{-1/2}, \quad y = G^{1/2} x.$$

Consequently every generalized eigenvalue is in F , and there is a basis x_j with $x_j^* G x_k = \delta_{jk}$. This is useful when the metric itself contains infinitesimal weights. It does not require a uniform ordinary-real lower bound on G . It does, however, use a square root of an arbitrary positive definite matrix, and so is one of the statements listed in [Remark 2.1](#) as needing a divisible workspace.

Every proof in this section used real closedness of F , either through the splitting of a characteristic polynomial or through a square root of an arbitrary positive element. [Section 9](#) proves the Hermitian and normal spectral theorems and the singular-value decomposition again, by a completely different route, over a Hahn workspace whose value group is *not* assumed divisible, and with control on which exponents can occur. Neither proof subsumes the other: the argument here is shorter and applies to any real closed field, while the argument there applies to any set-sized ordered abelian value group and returns support information that the algebraic argument cannot see.

5 Norms and variational principles with attained extrema

5.1 Operator and Frobenius norms

Define

$$\|A\|_{\text{op}} = \sigma_1(A), \quad \|A\|_{\text{F}} = \sqrt{\text{tr}(A^* A)} = \sqrt{\sum_{i,j} |a_{ij}|^2},$$

with operator norm zero for the zero matrix. The SVD proves, rather than assumes, that

$$\|A\|_{\text{op}} = \max_{\|x\|=1} \|Ax\|. \quad (5.1)$$

Indeed, writing $x = Vy$ gives $\|Ax\|^2 = \sum_j \sigma_j^2 |y_j|^2 \leq \sigma_1^2$, with equality at a leading right singular vector. Thus the symbol “max” is legitimate even though the sphere was never declared compact.

Both norms are invariant under compatible unitary multiplication. They satisfy triangle inequalities, and

$$\max_{i,j} |a_{ij}| \leq \|A\|_{\text{op}} \leq \|A\|_{\text{F}} \leq \sqrt{mn} \max_{i,j} |a_{ij}|, \quad (5.2)$$

$$\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}, \quad \|AB\|_{\text{F}} \leq \|A\|_{\text{op}} \|B\|_{\text{F}}. \quad (5.3)$$

For the first inequality in (5.2), apply A to a coordinate unit vector and then bound one coordinate by the vector norm. The middle inequality follows from $\sigma_1^2 \leq \sum_j \sigma_j^2$, and the last one from the finite sum defining the Frobenius norm. The operator product bound follows from (5.1); the Frobenius bound follows by applying it to every column of B and summing squares. The companion inequality $\|AB\|_{\text{F}} \leq \|A\|_{\text{F}} \|B\|_{\text{op}}$ follows by taking adjoints.

5.2 Courant–Fischer and Ky Fan principles

Theorem 5.1 (Variational principles). *Let $H = H^*$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$. For $1 \leq k \leq n$,*

$$\lambda_k = \max_{\dim W=k} \min_{\substack{x \in W \\ \|x\|=1}} x^* H x, \quad (5.4)$$

$$= \min_{\dim W=n-k+1} \max_{\substack{x \in W \\ \|x\|=1}} x^* H x. \quad (5.5)$$

Also, for $0 \leq k \leq n$,

$$\sum_{j=1}^k \lambda_j = \max_{X^* X = I_k} \operatorname{tr}(X^* H X). \quad (5.6)$$

All inner and outer extrema in these statements are attained.

Proof. For a fixed subspace W with orthonormal basis matrix X , the compression $X^* H X$ is Hermitian. Its largest and smallest eigenvalues are the extrema of the restricted Rayleigh quotient, by diagonalization. This proves inner attainment without compactness.

Every k -dimensional W meets the span of the eigenvectors for $\lambda_k, \dots, \lambda_n$ in a nonzero vector. After normalization that vector has Rayleigh quotient at most λ_k . The span of the first k eigenvectors attains the opposite bound, proving (5.4). The complementary dimension argument proves (5.5).

For (5.6), express the columns of X in an orthonormal eigenbasis of H . If a_j is the sum of their squared moduli in row j , then $0 \leq a_j \leq 1$ and $\sum_j a_j = k$. Hence

$$\operatorname{tr}(X^* H X) = \sum_j a_j \lambda_j \leq \sum_{j=1}^k \lambda_j.$$

To see the inequality even for signed λ_j , subtract λ_k from every λ_j . The terms before k have nonnegative coefficients and can only decrease on replacing 1 by a_j ; the terms after k are nonpositive. The constant part is unchanged because the weights sum to k . The first k eigenvectors attain equality. $\square$

Applying (5.4) to $A^* A$ and taking nonnegative square roots gives the singular-value principle

$$\sigma_k(A) = \max_{\dim W=k} \min_{\substack{x \in W \\ \|x\|=1}} \|Ax\|. \quad (5.7)$$

The list is padded with zeros to length n when necessary. Applying (5.6) to $-H$ similarly gives the minimum trace as the sum of the smallest k eigenvalues. These formulas are algebraic versions of familiar classical results [8]; their finite-dimensional proofs explain why they remain valid over F .

5.3 Perturbing eigenvalues and singular values

Theorem 5.2 (Ordered Lipschitz bounds). *If H, E are Hermitian of the same size, and both eigenvalue lists are ordered decreasingly, then*

$$|\lambda_j(H + E) - \lambda_j(H)| \leq \|E\|_{\text{op}}.$$

For arbitrary rectangular matrices of the same size,

$$|\sigma_j(A + E) - \sigma_j(A)| \leq \|E\|_{\text{op}}. \quad (5.8)$$

Proof. For a unit vector x , Cauchy–Schwarz gives $|x^*Ex| \leq \|E\|_{\text{op}}$. Insert the two inequalities for the perturbed Rayleigh quotient into (5.4), obtaining the Hermitian result. For singular values, the triangle inequality gives $|||(A + E)x| - |Ax||| \leq \|E\|_{\text{op}}$ on the unit sphere. Take the attained minima and maxima in (5.7). $\square$

Theorem 5.3 (A Frobenius eigenvalue bound). *For Hermitian H, G with decreasing eigenvalue lists λ, μ ,*

$$\sum_{j=1}^n (\lambda_j - \mu_j)^2 \leq \|H - G\|_{\text{F}}^2.$$

Proof. Let u_i, v_j be orthonormal eigenbases and set $p_{ij} = |u_i^* v_j|^2$. The matrix p has nonnegative entries and every row and column sum is one. Put $c_i = \sum_j p_{ij} \mu_j$. For every k ,

$$\sum_{i=1}^k c_i \leq \sum_{j=1}^k \mu_j,$$

by the weight argument in Theorem 5.1: the column weights $\sum_{i \leq k} p_{ij}$ lie in $[0, 1]$ and sum to k . Equality holds for $k = n$. Summation by parts, using $\lambda_i - \lambda_{i+1} \geq 0$, now gives

$$\text{tr}(HG) = \sum_i \lambda_i c_i \leq \sum_i \lambda_i \mu_i.$$

Expand $\|H - G\|_{\text{F}}^2 = \text{tr } H^2 + \text{tr } G^2 - 2 \text{tr}(HG)$ to conclude. $\square$

This proof does not use a decomposition of a doubly stochastic matrix into permutations, a measure on a unitary group, or a real-valued limiting argument. It is useful when all the eigenvalue movements, not just the largest one, matter.

6 Least squares, pseudoinverses, and best low-rank models

6.1 The Moore–Penrose inverse

Given an SVD $A = U\Sigma V^*$, let $\Sigma^\dagger$ be the transposed rectangular diagonal matrix whose nonzero entries are σ_j^{-1} and whose remaining entries are zero. Define

$$A^\dagger = V\Sigma^\dagger U^*.$$

Direct diagonal multiplication proves the Penrose identities

$$AA^\dagger A = A, \quad A^\dagger AA^\dagger = A^\dagger, \quad (AA^\dagger)^* = AA^\dagger, \quad (A^\dagger A)^* = A^\dagger A. \quad (6.1)$$

They characterize $A^\dagger$ uniquely, as in the classical construction [10]. For completeness, if B satisfies these identities, AB is the orthogonal projector onto $\text{im } A$: it is a Hermitian idempotent with that image. Similarly BA is the orthogonal projector onto $(\ker A)^\perp$, since $\ker(BA) = \ker A$. The equation $B = BAB$ then says that B is zero on $(\text{im } A)^\perp$ and, on $\text{im } A$, is the inverse of the restriction $A : (\ker A)^\perp \rightarrow \text{im } A$. This uniquely determines it and also shows that the SVD definition is basis-independent.

Theorem 6.1 (Least squares with an attained minimum). *For $b \in K^m$, all minimizers of $\|Ax - b\|$ are*

$$x = A^\dagger b + z, \quad z \in \ker A.$$

The unique minimizer of least norm is $A^\dagger b$, and the minimum residual is $\|(I - AA^\dagger)b\|$.

Proof. Write $c = U^*b$ and $y = V^*x$. The squared residual is

$$\sum_{j=1}^r |\sigma_j y_j - c_j|^2 + \sum_{j=r+1}^m |c_j|^2.$$

It is minimized precisely by $y_j = c_j/\sigma_j$ for $j \leq r$, with remaining input coordinates free. These free coordinates describe $\ker A$. They add a nonnegative squared norm to the fixed solution, so setting them to zero uniquely minimizes the norm. The residual formula is the corresponding orthogonal projection formula. $\square$

Small singular values are not discarded in this theorem. Their reciprocals may be infinite, and a mathematically exact least-squares solution may therefore have infinite norm even when the data are finite. Regularization in [Section 19](#) changes the optimization problem rather than silently changing what “nonzero” means.

6.2 Low-rank approximation

Let

$$A_k = \sum_{j=1}^k \sigma_j u_j v_j^*, \quad 0 \leq k < p.$$

Here zero terms can be included when $k > r$.

Theorem 6.2 (Eckart–Young bounds over F). *Among matrices B of rank at most k ,*

$$\min \|A - B\|_{\text{op}} = \sigma_{k+1}(A), \quad (6.2)$$

$$\min \|A - B\|_F^2 = \sum_{j=k+1}^p \sigma_j(A)^2. \quad (6.3)$$

Both minima are attained by A_k .

Proof. The SVD computes the two errors of A_k exactly. For the lower operator bound, a matrix B of rank at most k has a kernel meeting the span of $v_1, \dots, v_{k+1}$ in a nonzero vector. Normalize that vector x . Then

$$\|(A - B)x\| = \|Ax\| \geq \sigma_{k+1}.$$

For the Frobenius bound, choose $n - k$ orthonormal vectors w_j in $\ker B$ and extend them to an orthonormal basis. Summing squared column norms in that basis gives

$$\|A - B\|_F^2 \geq \sum_{j=1}^{n-k} \|Aw_j\|^2.$$

The minimum-trace version of (5.6), applied to A^*A , bounds the right side below by the sum of its smallest $n - k$ eigenvalues. That is exactly (6.3), after padding the singular list with zeros. $\square$

The least-squares lower-rank problem originates in Eckart and Young [9]. Here its optimal error is allowed to be infinitesimal or infinite. The theorem asserts existence of an optimizer, not uniqueness: repeated singular values and the operator norm in particular can produce many optimizers. It also gives a direct operational meaning to every singular scale: σ_{k+1} is the exact error required to replace the matrix by a rank- k model.

7 Hahn valuation and singular-value scales

From this section onward, unless a result explicitly returns to arbitrary F , work in (2.1). For $a \neq 0$, put

$$v(a) = \min \text{supp } a \in \Gamma, \quad \text{lc}(a) = a_{v(a)}.$$

Let

$$\mathcal{O} = \{a : v(a) \geq 0\}, \quad \mathfrak{m} = \{a : v(a) > 0\}.$$

The residue map $\text{st} : \mathcal{O} \rightarrow \mathbb{C}$ extracts the coefficient at exponent zero. On real finite elements it is the usual standard part. The valuation of a matrix or vector means the minimum valuation of its entries; all-zero objects have valuation ∞ . We sometimes write $v_{\text{mat}}(A)$ to make this convention explicit.

Remark 7.1 (Logical order). This section and the next use the singular-value decomposition. For a divisible Γ that is [Theorem 4.4](#). For an arbitrary set-sized ordered abelian Γ it is [Theorem 9.18](#), proved in [Section 9](#) by an argument that uses nothing from this section onward; likewise the positive square root taken inside [Lemma 7.2](#) is supplied without real closedness by [Lemma 9.8](#). So the scale theory below is available over an arbitrary Γ , and there is no circularity — only a forward reference, made so that the classical development can be read first.

7.1 Positivity prevents leading cancellation

Lemma 7.2 (Valuation of a Euclidean norm). *For any nonzero finite vector x and any nonzero matrix A ,*

$$v(\|x\|) = \min_j v(x_j), \quad v(\|A\|_{\text{op}}) = v(\|A\|_{\text{F}}) = \min_{i,j} v(a_{ij}). \quad (7.1)$$

Proof. Let $\alpha = \min_j v(x_j)$. The coefficient of $t^{2\alpha}$ in $\sum_j |x_j|^2$ is

$$\sum_{v(x_j)=\alpha} |\text{lc}(x_j)|^2 > 0.$$

Thus its valuation is 2α , and taking its positive square root gives valuation α . Apply the same argument to the entries of A for the Frobenius norm. Finally (5.2) traps the operator norm between a largest entry modulus and a fixed positive ordinary-real multiple of it. Both bounds have the same valuation, proving the remaining equality. $\square$

A finite sum of positive leading coefficients is crucial. An arbitrary sum of complex entries can cancel at its leading exponent; a squared Euclidean norm cannot.

Lemma 7.3 (Unitary matrices preserve the integral lattice). *If U is unitary, then $U, U^{-1} \in \mathcal{O}^{n \times n}$. Its reduction $\text{st } U$ is an ordinary complex unitary matrix, and $U\mathcal{O}^n = \mathcal{O}^n$.*

Proof. Every entry has modulus at most one because its column has squared norm one. Hence every entry has nonnegative valuation. The inverse is U^* and has the same property. Reducing $U^*U = I$ gives $(\text{st } U)^*(\text{st } U) = I$, and the two lattice inclusions follow from integrality of U and its inverse. $\square$

For the nonzero singular values of A , define

$$\gamma_j = v(\sigma_j), \quad \gamma_1 \leq \cdots \leq \gamma_r. \quad (7.2)$$

Their order is reversed because a larger positive Hahn number has a smaller or equal leading exponent. Different singular values may have the same valuation. The tuple (7.2) records only scales, not the positive leading coefficients or the rest of each expansion.

7.2 The determinantal characterization

Define the k th determinantal valuation by

$$\Delta_0(A) = 0, \quad \Delta_k(A) = \min_{\substack{I \subseteq \{1, \dots, m\}, |I|=k \\ J \subseteq \{1, \dots, n\}, |J|=k}} v(\det A_{I,J}), \quad 1 \leq k \leq p. \quad (7.3)$$

It is ∞ exactly when $k > r$.

Lemma 7.4 (Integral row and column invariance). *If $L \in \mathrm{GL}_m(\mathcal{O})$ and $R \in \mathrm{GL}_n(\mathcal{O})$, then $\Delta_k(LAR) = \Delta_k(A)$ for every k .*

Proof. By the finite Cauchy–Binet formula, every k -minor of LA is a sum of k -minors of A multiplied by k -minors of L . The latter coefficients are integral, so every summand has valuation at least $\Delta_k(A)$. The ultrametric inequality gives $\Delta_k(LA) \geq \Delta_k(A)$. Apply the same argument to L^{-1} for the reverse inequality. Right multiplication is identical, or follows by transposition. The argument also covers vanishing minors. $\square$

Theorem 7.5 (Singular scales are determinantal scales). *For a matrix A of rank r and $1 \leq k \leq r$,*

$$\Delta_k(A) = \gamma_1 + \dots + \gamma_k, \quad \gamma_k = \Delta_k(A) - \Delta_{k-1}(A). \quad (7.4)$$

In particular the finite slopes of the determinantal profile are nondecreasing. The tuple $(\gamma_1, \dots, \gamma_r)$ is a complete invariant for equivalence under integral invertible row and column operations, with the matrix dimensions fixed.

Proof. In an SVD $A = U\Sigma V^*$ both unitary factors and their inverses are integral by [Lemma 7.3](#). Thus [Lemma 7.4](#) reduces the calculation to Σ . Its nonzero k -minors are products of k distinct nonzero singular values. Their minimum valuation is the sum of the k smallest γ_j , proving (7.4).

Write $\sigma_j = t^{\gamma_j} u_j$, where $u_j \in \mathcal{O}^\times$ is a positive unit. Multiplying diagonal rows by u_j^{-1} shows that A is integrally equivalent to the rectangular diagonal matrix $\mathrm{diag}(t^{\gamma_1}, \dots, t^{\gamma_r}, 0, \dots)$. Thus two matrices with the same tuple are equivalent. Conversely, equivalence preserves Δ_k , hence the tuple and rank. $\square$

For integral matrices this is the singular-value interpretation of their valuation-ring invariant factors. The valuation ring need not be Noetherian or a discrete valuation ring. Only finitely many entries and minors are involved, so a minimum valuation among them exists. Unit equivalence remembers less than unitary equivalence: the latter preserves all singular values, not merely their valuations.

An exterior-power interpretation is also useful. Give $\bigwedge^k K^n$ the inner product for which wedges of coordinate basis vectors are orthonormal. The entries of $\bigwedge^k A$ are the k -minors of A . Wedges of the singular bases yield

$$\left\| \bigwedge^k A \right\|_{\mathrm{op}} = \sigma_1 \cdots \sigma_k.$$

Applying [Lemma 7.2](#) recovers (7.4). Thus a determinantal scale is the scale of the maximal k -dimensional volume factor.

7.3 Why entrywise scales alone are not enough

Example 7.6 (A nearly dependent matrix). Let $q \in \Gamma_{>0}$ and

$$A_q = \begin{pmatrix} 1 & 1 \\ 1 & 1 + t^q \end{pmatrix}.$$

Every entry has valuation zero. Nevertheless, $\det A_q = t^q$, so $\Delta_1 = 0$, $\Delta_2 = q$, and the singular scales are $0, q$. In contrast, the matrix of all ones has the same entrywise valuations and rank one. Changing q changes the hidden singular scale without changing any of those entrywise valuations.

The matrix A_q is real symmetric and positive definite, since its leading principal minors are 1 and t^q . Its singular values are its positive eigenvalues:

$$\sigma_{\pm} = \frac{2 + t^q \pm \sqrt{4 + t^{2q}}}{2}.$$

Hence $\text{lc}(\sigma_+) = 2$ and $\sigma_- = (1/2)t^q + \text{terms of valuation greater than } q$. A minimum-weight assignment using only entry valuations could miss the cancellation in the determinant. Exact minors or an equivalent elimination calculation are needed.

8 Gram coefficients recover scales and leading amplitudes

The polynomial report already develops Newton profiles of scalar polynomials [2]. Positive Gram matrices give an especially clean matrix input to that machinery: their characteristic coefficients carry no leading cancellation from squared minors.

Theorem 8.1 (Positive Cauchy–Binet identity). *For $A \in K^{m \times n}$ write*

$$\det(I_n + sA^*A) = \sum_{k=0}^p c_k s^k, \quad c_0 = 1. \quad (8.1)$$

Then

$$c_k = \sum_{|I|=|J|=k} |\det A_{I,J}|^2 = e_k(\sigma_1^2, \dots, \sigma_r^2). \quad (8.2)$$

For $1 \leq k \leq r$,

$$c_k > 0, \quad v(c_k) = 2\Delta_k(A). \quad (8.3)$$

For $k > r$, $c_k = 0$.

Proof. Expansion of $\det(I + sB)$ by the columns in which an entry of sB is selected shows that its coefficient of s^k is the sum of the principal k -minors of B . For $B = A^*A$, Cauchy–Binet expands the principal minor indexed by J as

$$\det((A^*A)_{J,J}) = \sum_{|I|=k} \overline{\det A_{I,J}} \det A_{I,J}.$$

This proves the first equality in (8.2). Diagonalizing A^*A gives the second. If a k -minor is nonzero, the leading coefficient of the sum of its squared moduli and all other squared minors of minimum valuation is a positive real number. Therefore no leading cancellation is possible, and the valuation is twice the minimum minor valuation. $\square$

Thus the Gram determinant polynomial provides all the singular scales without finding an eigenvector. Its nonzero coefficient valuations are $2\Delta_k$, with successive differences $2\gamma_k$. This statement works in any ordered group Γ ; it does not require drawing a polygon in an ordinary real plane.

8.1 Resolving singular values at a common scale

Define

$$C_k = \text{st}(t^{-2\Delta_k} c_k) > 0 \quad (0 \leq k \leq r), \quad C_0 = 1.$$

Let a maximal equal-scale block be

$$\gamma_{p_0+1} = \cdots = \gamma_{p_0+d} = \gamma,$$

where the preceding scales, if any, are strictly smaller and the following ones, if any, are strictly larger. Write $a_j = \text{lc}(\sigma_j) > 0$.

Theorem 8.2 (Residual polynomial for a singular-scale block). *For $0 \leq \ell \leq d$,*

$$\frac{C_{p_0+\ell}}{C_{p_0}} = e_\ell(a_{p_0+1}^2, \dots, a_{p_0+d}^2). \quad (8.4)$$

Consequently the ordinary real polynomial

$$R_\gamma(X) = X^d + \sum_{\ell=1}^d (-1)^\ell \frac{C_{p_0+\ell}}{C_{p_0}} X^{d-\ell} \quad (8.5)$$

has roots precisely the squared leading coefficients in that block, with multiplicity.

Proof. In the elementary symmetric sum for $c_{p_0+\ell}$, a product of minimum valuation must contain all the p_0 singular values of smaller valuation, exactly ℓ singular values from the block, and none from later blocks. Indeed, omitting a strictly smaller scale while using a larger one increases the sum of valuations; the increasing ordering shows that only choices across the equal boundary can tie the minimum. Therefore its leading coefficient is

$$\left(\prod_{j=1}^{p_0} a_j^2 \right) e_\ell(a_{p_0+1}^2, \dots, a_{p_0+d}^2).$$

For $\ell = 0$, the prefactor is exactly C_{p_0} . Division proves (8.4), and the elementary symmetric factorization proves (8.5). $\square$

This does not give all higher coefficients of the singular values. It gives their valuations and first nonzero coefficients; repeated roots of R_γ may require another scale refinement. It is already stronger than knowing only the Newton slopes.

For [Example 7.6](#),

$$c_1 = 4 + 2t^q + t^{2q}, \quad c_2 = t^{2q},$$

so $C_1 = 4$ and $C_2 = 1$. The two one-element blocks give leading squares 4 and 1/4, hence leading singular coefficients 2 and 1/2. For a same-scale example,

$$A = \text{diag}(2t^\gamma, 3t^\gamma), \quad C_1 = 13, \quad C_2 = 36,$$

and $R_\gamma(X) = X^2 - 13X + 36 = (X - 4)(X - 9)$. The valuation alone cannot distinguish the factors 2 and 3, but the positive Gram coefficients do.

8.2 The Gram case and its parity constraint

One consequence of [Theorem 7.5](#) will be used repeatedly below and is worth recording separately. It is a corollary, not a new identity: it is what [\(8.3\)](#) says once the determinantal scales of A^*A are themselves read off by [\(7.4\)](#).

Corollary 8.3 (Even determinantal scales of Gram matrices). *For $A \in K_\Gamma^{m \times n}$ and $1 \leq k \leq \text{rank}(A)$,*

$$\Delta_k(A^*A) = 2\Delta_k(A). \quad (8.6)$$

Proof. The nonzero eigenvalues of A^*A are the σ_j^2 , of valuations $2\gamma_j$ in the same nondecreasing order. Apply [\(7.4\)](#) to A^*A and to A . $\square$

Equivalently, the positive Cauchy–Binet identity prevents leading cancellation in the sums of squared minors, which is why [\(8.6\)](#) has no error term. The parity visible in it is an arithmetic property of the group Γ , not a statement about an integer-valued valuation; [Corollary 10.5](#) turns it into an exact criterion for factoring a positive matrix as a Gram matrix.

9 Support-certified descent without divisibility

Every decomposition in [Section 4](#) was obtained from real closedness of F , which [Section 2](#) supplies inside a Hahn workspace only when Γ is divisible. Passing to the divisible hull is harmless for existence and fatal for arithmetic: it makes every positive square root free, and so erases the distinction between a matrix whose spectral data need no new scales and one whose positive root genuinely does. This section keeps the exact exponent group. Throughout it, Γ is an arbitrary set-sized ordered abelian group and

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)) = F_\Gamma(i),$$

with no divisibility, finite-rank, countability or Archimedean assumption. Conjugation fixes every t^γ , and, as before, a positive exponent means a small monomial.

9.1 Strong summation is not a limit convention

Definition 9.1 (Strongly summable family). A set-indexed family $(f_i)_{i \in I}$ of Hahn series is *strongly summable* if $\bigcup_i \text{supp}(f_i)$ is well ordered and, for each exponent γ , only finitely many i have a nonzero coefficient at γ . Its sum is the coefficientwise finite sum.

This licenses arbitrary regrouping of a jointly summable family. It does not declare a sequence of partial sums to converge, and the distinction is not academic at higher rank.

Example 9.2 (A rank-two warning). Let $\Gamma = \mathbb{Z}^2$ with the lexicographic order, and put $\delta = (0, 1)$ and $\Omega = (1, 0)$, so that $0 < n\delta < \Omega$ for every positive integer n . The Hahn identity

$$\sum_{n \geq 0} t^{n\delta} = (1 - t^\delta)^{-1}$$

is valid, but its partial sums do not converge to the right-hand side in the valuation topology: the tail after N terms has valuation $(N+1)\delta < \Omega$ for every N . A remainder bound $v(\text{remainder}_N) \geq N\delta$ is therefore not by itself a convergence proof in arbitrary rank, and it is never used as one here. This is the same separation of algebraic identity, strong summability and workspace-relative topology drawn in [Section 18](#), exhibited in a value group where even the first two do not bring the third with them.

9.2 A self-contained support engine

Section 18 cites the positive-support lemma to Gonshor. Because finiteness *across all word lengths*, not merely at a fixed degree, is what the splitting construction needs, we prove it. The combinatorial method is the minimal-bad-sequence argument associated with Higman's lemma [18]; the resulting support statement is the Hahn–Neumann lemma [17].

Lemma 9.3 (Ordered words). *Let S be a well-ordered set, and order finite words in S by subsequence embedding with coordinatewise increase. Every infinite sequence of words has a pair $i < j$ such that the i th word embeds in the j th.*

Proof. Suppose a bad sequence exists. Choose it successively so that at each position the word has the least possible length among words allowing a bad continuation. No chosen word is empty. Write $w_i = u_i a_i$ with last letter a_i . A sequence in a well order has an infinite nondecreasing subsequence; choose $a_{i_0} \leq a_{i_1} \leq \dots$.

Then $w_0, \dots, w_{i_0-1}, u_{i_0}, u_{i_1}, \dots$ is still bad. A comparison from its initial segment into some u_{i_j} would compare the corresponding original words; and a comparison $u_{i_j} \preceq u_{i_k}$, together with $a_{i_j} \leq a_{i_k}$, would compare w_{i_j} with w_{i_k} . Both are impossible. This contradicts minimality of the length at position i_0 . $\square$

Lemma 9.4 (Positive support monoids). *Let $S \subseteq \Gamma_{>0}$ be well ordered. The additive monoid $\langle S \rangle_{\mathbb{N}}$ generated by S is well ordered, and each $\gamma \in \Gamma$ is the sum of only finitely many finite ordered words with letters in S .*

Proof. If the monoid admitted an infinite strictly decreasing sequence, choose a word representing each term. Lemma 9.3 gives an embedding of an earlier word into a later one; positivity of the unused letters and increase of the matched letters make the earlier sum at most the later one, a contradiction.

If infinitely many distinct words had sum γ , apply the same lemma to those words. Equality of sums forbids any unused positive letter and any strict increase of a matched letter, so the two words have the same length and the same letters, contrary to being distinct. $\square$

Lemma 9.5 (Hahn evaluation). *Let $z_1, \dots, z_d \in \mathfrak{m}$ and $S = \bigcup_{j=1}^d \text{supp}(z_j)$. Every formal series $f \in \mathbb{C}[[X_1, \dots, X_d]]$ has a well-defined strongly summed evaluation $f(z_1, \dots, z_d) \in K_{\Gamma}$, with support in $M = \langle S \rangle_{\mathbb{N}}$, and in $M \setminus \{0\}$ when $f(0) = 0$. Evaluation preserves sums, products, conjugation with real formal variables, and formal composition whose inner series have zero constant coefficient.*

Proof. A finite union of well-ordered subsets of a linear order is well ordered. Every contribution to a coefficient of $f(z)$ determines an ordered word of exponents in S together with a choice of one of the d variables at each position. By Lemma 9.4 there are, for a fixed exponent, only finitely many such words and labels, so every coefficient sum is finite and the support lies in M . For a product or a composition, expanding either side gives the same labelled words with the same finite contributions at each exponent, so regrouping is legal and the formal identities survive evaluation. Conjugation preserves supports, and the same coefficientwise argument applies. $\square$

Matrix formal series reduce to this lemma by taking finitely many scalar entries as the variables; the entries need not commute, since the scalar coefficient bookkeeping keeps all ordered matrix products apart. No bound on the growth of the ordinary complex Taylor coefficients is needed, because a coefficient at a fixed Hahn exponent is still a finite sum. This is exactly the mechanism behind Theorem 18.1, stated once in the generality the splitting construction requires.

Corollary 9.6 (Exponent-preserving inverses and unit roots). *If $h \in \mathfrak{m}$, the geometric and binomial series define $(1+h)^{-1}$ and $(1+h)^a$ for every rational a , and all their exponents lie in the group generated by $\text{supp}(h)$. Every nonzero Hahn series has an inverse without enlarging its exponent group.*

Proof. Apply the formal geometric and binomial identities and [Lemma 9.5](#). For a general series factor $f = ct^\alpha(1+h)$; its inverse is $c^{-1}t^{-\alpha}(1+h)^{-1}$, and both α and the exponents of h lie in the original support subgroup. $\square$

Lemma 9.7 (Formal maps are valuation nonexpanding). *Let $f \in \mathbb{C}[[X_1, \dots, X_d]]$ and $x, y \in \mathfrak{m}^d$. Then*

$$v(f(x) - f(y)) \geq \min_j v(x_j - y_j),$$

and the same entrywise assertion holds for a matrix of formal series.

Proof. There are formal series $g_j \in \mathbb{C}[[X, Y]]$ with $f(X) - f(Y) = \sum_j (X_j - Y_j)g_j(X, Y)$: for a monomial this is a telescoping product identity, and collecting by total degree gives the formal identity. Evaluate with [Lemma 9.5](#); each $g_j(x, y)$ is integral. $\square$

Lemma 9.8 (Norms exist without real closedness). *For every nonzero $x \in K_\Gamma^n$ we have $x^*x > 0$, and x^*x has a unique positive square root in F_Γ . The same holds for $\|A\|_\Gamma^2 = \text{tr}(A^*A)$ and for any finite sum of squared moduli of entries.*

Proof. Let $\alpha = v(x)$ and write $x = t^\alpha y$ with y integral and at least one entry of nonzero residue. The constant coefficient of y^*y is $\sum_j |\text{st}(y_j)|^2 > 0$, so $x^*x = ct^{2\alpha}(1+h)$ with $c > 0$ real and $h \in \mathfrak{m} \cap F_\Gamma$. The binomial series $(1+h)^{1/2}$ is a Hahn series by [Lemma 9.5](#), so $\sqrt{c}t^\alpha(1+h)^{1/2} \in F_\Gamma$ is the required root, and uniqueness follows from $(u-v)(u+v) = 0$ in an ordered field. $\square$

This is the one place where positivity does all the work. No real closedness is used, and none is available: if $\gamma \notin 2\Gamma$, the positive element t^γ has no square root in F_Γ and therefore cannot be a nonzero sum of squares. That single observation is why an SVD can stay in the base field while an arbitrary positive matrix square root need not, and it is quantified exactly in [Section 10](#). Note also that [Lemma 9.8](#) supplies the root whose existence [Lemma 7.2](#) takes for granted, so the valuation computation there is valid over an arbitrary Γ ; [Lemma 7.3](#) likewise used only integrality and is unaffected.

9.3 Separated-block lifting

Let

$$D = \text{diag}(c_1 I_{n_1}, \dots, c_s I_{n_s}), \quad c_j \in \mathbb{R}, \quad c_j \neq c_\ell \ (j \neq \ell), \quad (9.1)$$

and write Q_j for the corresponding constant coordinate projections. Suppose $E = E^* \in \mathfrak{m}^{n \times n}$ and $B = D + E$. Let S be the union of the supports of the entries of E and put $M = \langle S \rangle_\mathbb{N}$.

Theorem 9.9 (Separated-block lifting). *There are Hermitian projections $P_1, \dots, P_s$ and a unitary W over K_Γ such that*

$$P_j P_\ell = 0 \ (j \neq \ell), \quad \sum_j P_j = I, \quad B P_j = P_j B, \quad P_j \equiv Q_j \pmod{\mathfrak{m}}, \quad (9.2)$$

$$W \equiv I \pmod{\mathfrak{m}}, \quad P_j W = W Q_j, \quad W^* B W \text{ is block diagonal relative to } (Q_j). \quad (9.3)$$

Every entry of $P_j - Q_j$ and of $W - I$ is supported in $M \setminus \{0\}$. The construction uses universal finite-variable formal series in the entries of E , with ordinary constant coefficients.

Proof. Put $R_0(\zeta) = (\zeta I - D)^{-1}$ and define

$$P_j(E) = \sum_{k \geq 0} \text{Res}_{\zeta=c_j} \left[R_0(\zeta) (ER_0(\zeta))^k \right]. \quad (9.4)$$

For each k the residue is a homogeneous polynomial of degree k in the entries of E whose coefficients are ordinary complex numbers, obtained as residues of rational functions with poles among the fixed real c_j . Hence Lemma 9.5 makes (9.4) a strongly summed Hahn series with the stated support; the term $k = 0$ is Q_j .

The projector identities are justified without presupposing a spectral theorem over K_Γ . Parametrize Hermitian E by n^2 independent *ordinary real* coordinates. For sufficiently small such coordinates, disjoint complex circles around the c_j separate the eigenvalues of the ordinary complex Hermitian matrix $D + E$: if λ is farther than $\|E\|$ from every c_j , then $I - (\lambda I - D)^{-1}E$ is invertible by its ordinary geometric series, so $\lambda I - D - E$ is invertible. On those circles the ordinary resolvent has its Neumann expansion, and the contour integrals are the orthogonal spectral projectors, as one sees by diagonalizing the ordinary Hermitian matrix. Their Taylor series at $E = 0$ are exactly (9.4). Therefore $P_j^* = P_j$, $P_j^2 = P_j$, $P_j P_\ell = 0$, $\sum_j P_j = I$ and $[B, P_j] = 0$ hold as finite-variable formal identities, every Taylor coefficient of the difference being zero, and Hahn evaluation transports them to our E . This is a proof of universal coefficient identities by ordinary complex analysis, *not* contour integration in a surreal topology.

Now set

$$X = \sum_j P_j Q_j, \quad G = X^* X = \sum_j Q_j P_j Q_j, \quad W = X G^{-1/2}. \quad (9.5)$$

Here $X \in I + \mathfrak{m}^{n \times n}$ and $G = I + H$ with $H \in \mathfrak{m}^{n \times n}$, and

$$G^{-1/2} = \sum_{k \geq 0} \binom{-1/2}{k} H^k \quad (9.6)$$

is strongly summable by Lemma 9.5. It is Hermitian, commutes with G and squares to G^{-1} , so $W^* W = I$; a square matrix with a left inverse is invertible, so also $W W^* = I$. Orthogonality of the projectors gives $P_j X = X Q_j$, while G , and hence $G^{-1/2}$, commutes with every Q_j ; thus $P_j W = W Q_j$. Since B commutes with each P_j , its conjugate $W^* B W$ commutes with every Q_j , which is block diagonality. All nonconstant terms are formal series without constant term in the entries of E , so their supports lie in $M \setminus \{0\}$. $\square$

Remark 9.10 (What is canonical and what is not). For a fixed residue decomposition D and projections Q_j , the formulas (9.4) and (9.5) specify a particular lift. They do not choose a preferred ordinary eigenbasis inside a repeated eigenspace of a later leading coefficient; those finite-dimensional choices are recorded in the splitting tree below. This is the same distinction as the one after (4.1): the eigenspace projector is canonical, a basis within it is not.

Writing $E_{j\ell} = Q_j E Q_\ell$ and taking the $k = 1$ residues,

$$W = I + K + O_{\text{formal}}(E^2), \quad K_{j\ell} = \frac{E_{j\ell}}{c_\ell - c_j} \ (j \neq \ell), \quad K_{jj} = 0, \quad (9.7)$$

with K skew-Hermitian; $X = I + K + O(E^2)$ gives $X^* X = I + O(E^2)$. The first-order off-diagonal term of $W^*(D + E)W$ is $E + [D, K]$, which vanishes between different blocks, and the diagonal block to second degree is

$$c_j I + E_{jj} + \sum_{\ell \neq j} \frac{E_{j\ell} E_{\ell j}}{c_j - c_\ell} + O_{\text{formal}}(E^3). \quad (9.8)$$

These O_{formal} symbols mean support-certified sums of monomials of the indicated total degrees, not a topological limiting convention. The second-order coupling correction in (9.8) is the formal-series counterpart of the Schur-complement correction (17.2), and Example 17.2 is the same phenomenon at $n = 2$.

Proposition 9.11 (Local valuation estimate). *Apply the construction with the same D, Q_j to Hermitian $E, E' \in \mathfrak{m}^{n \times n}$. Then*

$$v(P_j(E) - P_j(E')) \geq v(E - E'), \quad v(W(E) - W(E')) \geq v(E - E').$$

If $A = \mu I + t^\alpha V(D + E)V^*$ with V constant unitary, and $v(A' - A) > \alpha$, then the corresponding cluster projectors satisfy

$$v(\Pi_j(A') - \Pi_j(A)) \geq v(A' - A) - \alpha. \quad (9.9)$$

Proof. The first pair of inequalities is Lemma 9.7 applied to the universal formal entries. For the second, keep the same μ, α, V and put $E' = E + t^{-\alpha}V^*(A' - A)V$; its entries are infinitesimal, and Lemma 7.3 gives $v(E' - E) = v(A' - A) - \alpha$. $\square$

The estimate is sharp already for a two-by-two constant diagonal matrix perturbed off the diagonal, since (9.7) has a nonzero linear term. Its role is to give a support-compatible *valuation* form of separated-cluster stability. It does not supersede Theorem 16.1 and Corollary 16.2, which measure an ordered Frobenius displacement against an ordered gap; the two are different statements, and Example 16.3 shows that an infinitesimal valuation hypothesis alone cannot control an ordered subspace rotation when the gap is infinitesimal too.

9.4 Finite splitting trees and exact spectral descent

Theorem 9.12 (Hermitian Hahn spectral theorem). *Let Γ be any set-sized ordered abelian group. For every Hermitian $A \in K_\Gamma^{n \times n}$ there are a unitary U over K_Γ and $\lambda_1, \dots, \lambda_n \in F_\Gamma$ with $U^*AU = \text{diag}(\lambda_1, \dots, \lambda_n)$. For real symmetric input U can be chosen real orthogonal. Every exponent occurring in U or in the λ_j lies in the subgroup generated by the exponents of the input entries.*

Proof. Induct on n , using no algebraic closedness or real closedness assumption. The case $n = 1$ is immediate. Set

$$\mu = \frac{1}{n} \text{tr}(A), \quad C = A - \mu I. \quad (9.10)$$

If $C = 0$ take $U = I$. Otherwise let $\alpha = v(C)$ and write

$$t^{-\alpha}C = C_0 + E_0, \quad C_0 \in \mathbb{C}^{n \times n}, \quad E_0 \in \mathfrak{m}^{n \times n}. \quad (9.11)$$

The constant matrix C_0 is nonzero, Hermitian and traceless, so it has at least two distinct ordinary real eigenvalues: by the ordinary spectral theorem a Hermitian matrix with one eigenvalue is scalar, and a traceless scalar matrix in characteristic zero is zero. Choose an ordinary constant unitary V with $V^*C_0V = D$ of the form (9.1), with $s \geq 2$, and apply Theorem 9.9 to $D + V^*E_0V$. Then $W^*V^*AVW = \text{diag}(A_1, \dots, A_s)$ with each A_j Hermitian of strictly smaller size. By induction choose unitary U_j diagonalizing A_j ; then $U = VW \text{diag}(U_1, \dots, U_s)$ diagonalizes A , and the diagonal entries are real because the diagonal matrix is Hermitian.

For the support claim let H be the subgroup generated by the input supports. Both μ and C have supports in H , and $\alpha \in H$, so normalization only translates supports inside H ; constant changes of basis add no exponents; and Theorem 9.9 uses sums of positive exponents already in H , so W and the A_j stay in K_H . The induction therefore runs entirely inside K_H . For real input use real constant eigenbases; every coefficient of the projector and polar constructions is then real. $\square$

This is [Theorem 4.1](#) again with two hypotheses removed and one conclusion added. It does not replace the earlier proof, which is three lines long and applies to any real closed field; it is the version that survives when K_Γ is not algebraically closed, and it alone says where the exponents can be.

Corollary 9.13 (The splitting tree is finite). *The construction in [Theorem 9.12](#) has at most $n - 1$ nontrivial splitting nodes. This bound is independent of the order type of the input supports and of the rank or cardinality of Γ .*

Proof. Every internal node has at least two children, and the dimension at a node is the sum of its children's positive dimensions, so there are at most n leaves. A finite rooted tree in which every internal node has at least two children has at most one fewer internal node than it has leaves. Termination follows from the strictly decreasing positive dimensions along each branch. $\square$

At each internal node the construction records the scalar series μ , the exact leading exponent α , the ordinary constant spectral data (V, D, Q_j) , and the well-ordered positive support S of the normalized perturbation, followed by the inclusion of the lifted projector and unitary supports in $\langle S \rangle_{\mathbb{N}}$ and then the certificates of the smaller blocks. A scalar leaf records its series and multiplicity.

This is a *finite tree of set-valued support data*, not in general a finite list of monomials: a single node can carry an infinite or uncountable well-ordered support. Nor is it automatically an effective algorithm from an arbitrary presentation of a surreal number. Exact zero tests, support access and extraction of a leading coefficient must be supplied by the representation — the same contract as in [Section 13](#), now for a harder task. The coefficient fields remain the full $\mathbb{R}$ and $\mathbb{C}$; preservation of a smaller coefficient field such as $\mathbb{Q}$ is not asserted.

Example 9.14 (A scalar tail beyond every finite truncation). In the rank-two group of [Example 9.2](#), put

$$c = \sum_{m \geq 1} t^{m\delta}, \quad A = cI + t^\Omega \operatorname{diag}(1, -1).$$

Every coefficient encountered at the scales $m\delta$ is scalar, so a procedure that inspects more and more of those coefficients does not reach the first nonscalar scale after any finite number of inspections. The exact trace subtraction gives $A - \frac{1}{2} \operatorname{tr}(A)I = t^\Omega \operatorname{diag}(1, -1)$ at once. This is why [\(9.10\)](#) is stated as an exact operation, and it is the spectral counterpart of the truncation failure in [Section 13](#).

Corollary 9.15 (Normal spectral descent). *Every normal matrix over K_Γ is unitarily diagonalizable over K_Γ , without an exponent-group extension.*

Proof. Write $A = H + iJ$ with $H = (A + A^*)/2$ and $J = (A - A^*)/(2i)$ Hermitian; normality is equivalent to $HJ = JH$. Diagonalize H by [Theorem 9.12](#). The relation $HJ = JH$ makes the entries of J between distinct exact eigenspaces of H vanish, so diagonalize the Hermitian restriction of J inside each block. The resulting unitary diagonalizes H and J , hence A , and every operation stayed in the original support group. $\square$

[Proposition 4.2](#) proves more about general matrices — a Schur form for every square matrix — but it needs an eigenvalue to exist, hence an algebraically closed K . [Corollary 9.15](#) covers only normal matrices and needs nothing of the sort. Neither contains the other.

Lemma 9.16 (A finite-dimensional commutant fact). *If A is normal over K_Γ and $BA = AB$, then $BA^* = A^*B$.*

Proof. Use [Corollary 9.15](#) to make A diagonal. The entry b_{jk} must vanish when $\lambda_j \neq \lambda_k$, and these are exactly the pairs with $\overline{\lambda_j} \neq \overline{\lambda_k}$; so B also commutes with the diagonal conjugate of A . $\square$

Theorem 9.17 (Finite-witness simultaneous diagonalization). *Let $\mathcal{F} \subseteq K_\Gamma^{n \times n}$ be a set of pairwise commuting normal matrices. There is a simultaneous unitary diagonalizer, and it can be constructed from at most $n - 1$ selected members of $\mathcal{F}$; consequently its support may be confined to the subgroup generated by those finitely many members' supports. After selection, one ordinary complex linear combination of the selected matrices can be chosen whose eigenspaces are exactly the final joint eigenspaces.*

Proof. Begin with the whole space as one block. If every member acts as a scalar there, stop. Otherwise choose a nonscalar member and decompose the block into its distinct eigenspaces by [Corollary 9.15](#). Every other member and its adjoint preserve those spaces by [Lemma 9.16](#), so their restrictions remain normal; repeat on any block on which some member is nonscalar. Each selection strictly increases the number of joint blocks, which never exceeds n , so at most $n - 1$ selections are needed, and the successive diagonalizers are produced from the selected matrices and previously constructed bases only.

Let $B_1, \dots, B_m$ be the selected matrices and let their joint scalar tuples on the final blocks be $\lambda^{(a)} \in K_\Gamma^m$; these are pairwise distinct. For each pair $a \neq b$ the condition $\sum_j c_j (\lambda_j^{(a)} - \lambda_j^{(b)}) = 0$ defines a proper complex-linear subspace of $\mathbb{C}^m$, and a finite union of proper subspaces over the infinite field $\mathbb{C}$ does not fill $\mathbb{C}^m$. Choose (c_j) outside that union; then $\sum_j c_j B_j$ has distinct scalar values on the final blocks, and it is normal because the selected matrices commute with one another and with one another's adjoints. $\square$

For a family of scalar matrices no member need be selected. This is an existence theorem for a set-sized family, not a procedure for searching an arbitrary undecidable one.

9.5 Singular values stay in the original scale group

Theorem 9.18 (SVD over a nondivisible Hahn field). *Every $A \in K_\Gamma^{m \times n}$ admits $A = U\Sigma V^*$ with U, V unitary over K_Γ and Σ rectangular diagonal with entries $\sigma_1 \geq \dots \geq \sigma_r > 0$ in F_Γ and zeros elsewhere. The factors can be chosen with every exponent in the subgroup generated by the input supports.*

Proof. Diagonalize A^*A by [Theorem 9.12](#). If v_j is a unit eigenvector with eigenvalue λ_j , then $\lambda_j = (Av_j)^*(Av_j) \geq 0$. When $\lambda_j > 0$, [Lemma 9.8](#) gives $\sigma_j = \sqrt{\lambda_j} \in F_\Gamma$ with no extension of the group; put $u_j = Av_j/\sigma_j$, which are orthonormal since $u_j^*u_k = v_j^*A^*Av_k/(\sigma_j\sigma_k) = \delta_{jk}$. When $\lambda_j = 0$, [Lemma 9.8](#) forces $Av_j = 0$. Extend the nonzero u_j to an orthonormal basis by finite Gram–Schmidt, selecting standard basis vectors not yet in the span; every nonzero residual vector has its norm in F_Γ by [Lemma 9.8](#), so no step needs a square root of an arbitrary positive scalar. Reordering the finitely many positive singular values gives the stated form and preserves the support field. $\square$

Corollary 9.19 (Gram square roots and polar decomposition). *For every rectangular A over K_Γ , the Gram matrix A^*A has its principal positive semidefinite square root over K_Γ ; in particular $|A|$ and the polar factorization (4.2) are defined over K_Γ with no enlargement of the value group. For square invertible A , the polar decomposition $A = QP$ has Q unitary and P positive definite over K_Γ .*

Proof. With the notation of [Theorem 9.18](#) put $P = V \operatorname{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)V^*$, so $P^2 = A^*A$. In the invertible square case take $Q = AP^{-1}$, or $Q = UV^*$. $\square$

This does *not* say that every positive semidefinite matrix is a Gram matrix over a nondivisible base field. The scalar matrix (t^γ) with $\gamma \notin 2\Gamma$ is a counterexample, and [Corollary 10.5](#) gives the exact characterization. Note also that [Theorem 7.5](#) and [Theorem 8.1](#) are now available over an arbitrary Γ : their proofs used only the SVD, integrality of unitary factors ([Lemma 7.3](#)), Cauchy–Binet and positivity of a sum of squared minors, and [Theorem 9.18](#) supplies the first without divisibility.

9.6 The subgroup, and what it is not

Three sharp examples fix the boundaries of the descent statement.

Example 9.20 (New integer combinations, but no new fractions). For $\Gamma = \mathbb{Z}$ and $t = t^1$, take

$$A = \begin{pmatrix} t^3 & t^4 \\ t^4 & 0 \end{pmatrix}, \quad \lambda_{\pm} = \frac{t^3}{2}(1 \pm \sqrt{1 + 4t^2}), \quad (9.12)$$

so that $\lambda_+ = t^3 + t^5 - t^7 + 2t^9 - \dots$ and $\lambda_- = -t^5 + t^7 - 2t^9 + \dots$. With $w = (\sqrt{1 + 4t^2} - 1)/(2t) = t - t^3 + 2t^5 - \dots$, the orthogonal matrix $U = (1 + w^2)^{-1/2} \begin{pmatrix} 1 & -w \\ w & 1 \end{pmatrix}$ satisfies $U^*AU = \text{diag}(\lambda_+, \lambda_-)$, and all exponents are integers. But the exponent 5 is *not* in the nonnegative monoid generated by the raw input exponents 3 and 4. A theorem asserting that eigenvalue supports stay in the raw input monoid would therefore be false. The correct invariant is the input *subgroup*, supplemented by the positive monoids of the normalized perturbations at each node.

Example 9.21 (An eigenvalue beyond all ordinary perturbation orders). In the rank-two group of [Example 9.2](#) write $t = t^\delta$ and $u = t^\Omega$, and let $A = \begin{pmatrix} 0 & u \\ u & t \end{pmatrix}$. Since $v(u/t) = \Omega - \delta > 0$, the smaller eigenvalue is

$$\lambda_- = \frac{t}{2}(1 - \sqrt{1 + 4u^2/t^2}) = -u^2t^{-1} + u^4t^{-3} - 2u^6t^{-5} + \dots,$$

of valuation $2\Omega - \delta$, which exceeds every finite multiple of δ . The support stays in $\mathbb{Z}\delta + \mathbb{Z}\Omega$ and the expansion is certified by the positive generator $2(\Omega - \delta)$. A finite-order expansion measured only in powers of t never sees this eigenvalue; compare [Example 12.3](#), where no finite integer cutoff reaches the last singular direction.

Proposition 9.22 (Order-preserving transport). *Let $\phi : \Gamma \hookrightarrow \Lambda$ be an injective order-preserving group homomorphism. Then $\phi_*(\sum_\gamma a_\gamma t^\gamma) = \sum_\gamma a_\gamma s^{\phi(\gamma)}$ is an injective star-preserving field homomorphism, and it preserves strongly summable families, the support certificates of the splitting construction, and each chosen spectral or singular-value decomposition.*

Proof. An order embedding sends a well-ordered support to a well-ordered support, and it preserves and reflects equality of sums of exponents, so the finite coefficient formulas for strong sums and products are unchanged. It preserves leading coefficients and minima. Hence the trace subtraction, the normalization, the formal projectors, the polar intertwiner and all selected ordinary constant eigenbases transport through every node of the finite tree. $\square$

This lets one symbolic support construction be interpreted at different surreal scales with the same ordered-group relations. It does not claim that arbitrary numerical substitutions, in particular ones that identify or reverse exponents, preserve Hahn summability.

Warning 9.23 (Relative degrees can drop in a larger workspace). Under an embedding into a larger group, an exponent not divisible by q in the source may become divisible in the target: $\mathbb{Z} \hookrightarrow \mathbb{Q}$ makes every positive monomial's q th root available and kills all ramification. The spectral identities transport by [Proposition 9.22](#), but the relative ramification group of [Section 10](#) must be recomputed in the new base group. Degree is preserved under an isomorphism onto the image group, *not* under arbitrary enlargement of the base workspace.

Warning 9.24 (This is not a multivariable analytic eigenbasis theorem). Consider the real symmetric formal matrix

$$H(x, y) = \begin{pmatrix} x & y \\ y & -x \end{pmatrix}, \quad (9.13)$$

whose characteristic equation is $\lambda^2 = x^2 + y^2$. There is no solution $\lambda \in \mathbb{C}[[x, y]]$: its lowest homogeneous term would have to be $ax + by$ with $a^2 = b^2 = 1$ and $2ab = 0$. So a general eigenvalue or eigenbasis theorem in that formal power-series ring is false [15, 16]. After the ordered Hahn specialization $x = t^\alpha$, $y = t^\beta$ with $\beta > \alpha$, by contrast,

$$\sqrt{x^2 + y^2} = t^\alpha \sum_{m \geq 0} \binom{1/2}{m} t^{2m(\beta - \alpha)}$$

is a legitimate Hahn series in the original group, because the ordering permits division by the dominant monomial. Its displayed support may correspond to negative powers of x in the original two-variable expression, so it gives neither an element of $\mathbb{C}[[x, y]]$, nor a common analytic germ through the crossing, nor a global analytic choice of eigenvectors. The descent theorems resolve the totally ordered scale problem, and only that one.

10 Determinantal ramification and the exact root field

Spectral data need no new scales. Positive matrix roots sometimes do, and the cost is exactly computable from the invariant already in hand. Every ordered abelian group is torsion free, so its divisible hull $\Gamma_{\mathbb{Q}} = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$ carries a unique order extending that of Γ ; we write γ/q only as an element of this hull unless its membership in a smaller group has been established. Throughout this section and the next, Γ is again arbitrary.

10.1 Scalar roots are monomials times base-field units

Lemma 10.1 (The scalar root formula). *Let $a \in F_{\Gamma}$ be positive, let $\gamma = v(a)$, and let $q \geq 2$ be an integer. There is a unique positive q th root in $F_{\Gamma_{\mathbb{Q}}}$, and it has the form*

$$a^{1/q} = t^{\gamma/q} u, \quad u \in F_{\Gamma}, \quad u > 0. \quad (10.1)$$

It belongs to F_{Γ} if and only if $\gamma \in q\Gamma$.

Proof. Factor $a = ct^{\gamma}(1 + h)$ with $c > 0$ real and $h \in \mathfrak{m} \cap F_{\Gamma}$. The binomial series gives $u = c^{1/q}(1 + h)^{1/q} \in F_{\Gamma}$ by Corollary 9.6, with positive constant coefficient, proving existence and the displayed factorization. The map $x \mapsto x^q$ is strictly increasing on positive elements of any ordered field, so the positive root is unique. A root in F_{Γ} must have valuation $\gamma/q \in \Gamma$, and the formula proves the converse. $\square$

Lemma 10.2 (Finite-index Hahn extensions). *Let $\Gamma \subseteq \Lambda \subseteq \Gamma_{\mathbb{Q}}$ with Λ/Γ finite. Then*

$$[K_{\Lambda} : K_{\Gamma}] = [F_{\Lambda} : F_{\Gamma}] = |\Lambda/\Gamma|, \quad (10.2)$$

and a choice of coset representatives η_h gives a basis $(t^{\eta_h})_{h \in \Lambda/\Gamma}$ over the respective base field. If $q\Lambda \subseteq \Gamma$, the complex extension is Galois and

$$\text{Gal}(K_{\Lambda}/K_{\Gamma}) = \text{Hom}(\Lambda/\Gamma, \mu_q), \quad (10.3)$$

where $\mu_q \subset \mathbb{C}^{\times}$ is the group of q th roots of unity and a character χ acts by $t^{\eta} \mapsto \chi(\eta + \Gamma)t^{\eta}$. The extension is totally ramified: its residue field is still $\mathbb{C}$ and its ramification index is the full degree.

Proof. Split the support of a Hahn series into the finitely many cosets of Γ . Translating the part in coset h by $-\eta_h$ gives a well-ordered subset of Γ , so every series has a unique expression $f = \sum_h t^{\eta_h} f_h$ with $f_h \in K_\Gamma$, and the analogous real expression; uniqueness holds because supports in distinct cosets are disjoint. This proves the basis and the degree.

Multiplication by a character on each coset preserves all coefficientwise sums and products, so it defines an automorphism. A finite abelian group killed by q has exactly its own order many characters into μ_q , and these separate points, as one sees by decomposing it into cyclic factors. We have thus exhibited as many distinct automorphisms as the degree, so the extension is Galois with the stated group. The residue field is unchanged and the value group grows by the full index, which is total ramification. $\square$

The finite-index hypothesis is essential, not cosmetic. For an infinite group extension, the field generated by all the new monomials need not contain every Hahn series supported in the enlarged group. We invoke [Lemma 10.2](#) only for extensions generated by finitely many fractional exponents of bounded denominator.

10.2 Existence, uniqueness, and the exact extension

Theorem 10.3 (Principal-root field). *Let $A \in K_\Gamma^{n \times n}$ be Hermitian positive semidefinite, with positive eigenvalues $\lambda_1, \dots, \lambda_r$ counted with multiplicity, and put $\gamma_j = v(\lambda_j)$ and $\Lambda = \Gamma + \sum_{j=1}^r \mathbb{Z}\gamma_j/q$. There is a unique positive semidefinite Hermitian B over $K_{\Gamma_\mathbb{Q}}$ with $B^q = A$. It lies in $K_\Lambda^{n \times n}$, and*

$$K_\Gamma(B) = K_\Lambda, \quad (10.4)$$

where $K_\Gamma(B)$ denotes the field obtained by adjoining all entries of B .

Proof. Choose a unitary diagonalization over the base field, furnished by [Theorem 9.12](#): $A = U \operatorname{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0) U^*$. The diagonal entries are nonnegative, being values of the quadratic form on unit eigenvectors. By [Lemma 10.1](#), $b_j = \lambda_j^{1/q} = t^{\gamma_j/q} u_j$ with $u_j \in F_\Gamma$ positive. Set

$$B = U \operatorname{diag}(b_1, \dots, b_r, 0, \dots, 0) U^*, \quad (10.5)$$

which is positive semidefinite, satisfies $B^q = A$, and has entries in K_Λ .

For uniqueness let $C = C^* \succeq 0$ satisfy $C^q = A$ in the enlarged field. Then C and A commute, so [Theorem 9.17](#) gives a simultaneous Hermitian diagonalization there. On an eigenspace of A with value λ , every eigenvalue of C is nonnegative with q th power λ , and the ordered scalar uniqueness of [Lemma 10.1](#) forces it to be the chosen root, including 0 when $\lambda = 0$. Hence $C = B$.

Let $L = K_\Gamma(B)$. Since U lies in the base field, the entries of $U^* B U$ show every $b_j \in L$; dividing by the base-field unit u_j gives $t^{\gamma_j/q} \in L$. These finitely many monomials, together with monomials already in the base field, generate all coset representatives of Λ/Γ by multiplication and inversion, and the quotient is finite, being generated by r elements killed by q . The basis in [Lemma 10.2](#) therefore gives $K_\Lambda \subseteq L$, and the reverse inclusion is (10.5). $\square$

This is the exact sense in which [Proposition 4.3](#) needs a divisible workspace. The positive semidefinite root exists and is unique, but over a general Γ it lives in K_Λ , and Λ can be strictly larger than Γ .

10.3 Replacing eigenvalues by minors

The eigenvalues in [Theorem 10.3](#) can be eliminated entirely. By (7.4) the determinantal valuations and the eigenvalue valuations determine one another, so the field of definition is computable from finitely many minors and no spectral decomposition at all.

Theorem 10.4 (Determinantal ramification law). *Let $A \in K_\Gamma^{n \times n}$ be Hermitian positive semidefinite of rank r , let $\Delta_k = \Delta_k(A)$ be as in (7.3), and for an integer $q \geq 2$ put*

$$\Gamma_{A,q} = \Gamma + \sum_{k=1}^r \mathbb{Z} \frac{\Delta_k}{q} \subseteq \Gamma_{\mathbb{Q}}, \quad R_q(A) = \langle \Delta_1 + q\Gamma, \dots, \Delta_r + q\Gamma \rangle \subseteq \Gamma/q\Gamma. \quad (10.6)$$

Then the principal positive semidefinite root satisfies

$$K_\Gamma(A^{1/q}) = K_{\Gamma_{A,q}}, \quad (10.7)$$

$$[K_\Gamma(A^{1/q}) : K_\Gamma] = |R_q(A)| \leq q^r. \quad (10.8)$$

In particular

$$A^{1/q} \in K_\Gamma^{n \times n} \iff \Delta_k(A) \in q\Gamma \quad (1 \leq k \leq r). \quad (10.9)$$

The extension is finite, abelian and totally ramified, with Galois group isomorphic to $\text{Hom}(R_q(A), \mu_q)$.

Proof. Order the γ_j nondecreasingly. By (7.4), $\Delta_k = \gamma_1 + \dots + \gamma_k$, so

$$\Gamma + \sum_j \mathbb{Z} \frac{\gamma_j}{q} = \Gamma + \sum_k \mathbb{Z} \frac{\Delta_k}{q} : \quad (10.10)$$

each Δ_k/q is a sum of the γ_j/q , and each $\gamma_j/q = (\Delta_j - \Delta_{j-1})/q$. The map $\Gamma/q\Gamma \rightarrow (q^{-1}\Gamma)/\Gamma$ sending $\gamma + q\Gamma$ to $\gamma/q + \Gamma$ is an isomorphism, and it identifies $R_q(A)$ with $\Gamma_{A,q}/\Gamma$. Combining Theorem 10.3 with Lemma 10.2 gives (10.7), (10.8) and the Galois assertion, and the extension is trivial exactly when every generating class vanishes, which is (10.9). $\square$

Two things deserve emphasis. First, (10.8) is not merely a bound: it computes the exact degree from finitely many minor valuations, without computing an eigenbasis or the eigenvalues. Second, the same invariant Δ_k that Theorem 7.5 uses to record singular scales here controls a finite radical extension; the determinantal profile is doing two different jobs, and (10.9) is the second one in its sharpest form.

Corollary 10.5 (Exact Gram-factorization criterion). *A Hermitian positive semidefinite A over K_Γ can be written $A = C^*C$ with C over K_Γ if and only if $\Delta_k(A) \in 2\Gamma$ for $1 \leq k \leq \text{rank}(A)$. The condition permits C to be square, and it is necessary for every rectangular Gram representation.*

Proof. Necessity is Corollary 8.3: if $A = C^*C$ then $\Delta_k(A) = 2\Delta_k(C) \in 2\Gamma$. For sufficiency, Theorem 10.4 with $q = 2$ supplies the principal square root B over K_Γ ; take $C = B$. $\square$

Corollary 10.6 (Rational powers). *Let A be positive definite, $p \in \mathbb{Z}$, $q \geq 2$, $\gcd(p, q) = 1$. Then the principal power $A^{p/q}$ generates the same extension as $A^{1/q}$. For positive semidefinite A the same holds when $p > 0$.*

Proof. The nonzero diagonal spectral entries of $A^{p/q}$ are b_j^p . If $ap + bq = 1$, then $b_j = (b_j^p)^a \lambda_j^b$, so adjoining these entries is field-theoretically equivalent to adjoining the b_j . Zero eigenvalues are omitted, and negative powers require invertibility. $\square$

10.4 Prime and composite denominators

When q is prime, $\Gamma/q\Gamma$ is a vector space over $\mathbf{F}_q$ and

$$[K_\Gamma(A^{1/q}) : K_\Gamma] = q^{\dim_{\mathbf{F}_q} \text{span}\{\Delta_k + q\Gamma : 1 \leq k \leq r\}}.$$

For composite q the exact invariant is the finite subgroup, not the rank of a vector space: a single class of order 2 modulo 4Γ contributes a quadratic extension, not a quartic one.

If $\Gamma = \mathbb{Z}^d$ with any fixed compatible order and the Δ_k are integer vectors, put $L = q\mathbb{Z}^d + \sum_{k=1}^r \mathbb{Z}\Delta_k \subseteq \mathbb{Z}^d$. Then

$$|R_q(A)| = \frac{q^d}{[\mathbb{Z}^d : L]}, \quad (10.11)$$

and the index is computed by the Smith normal form of the integer matrix with columns $qI_d, \Delta_1, \dots, \Delta_r$. The root-field degree is therefore a finite integer-lattice computation once the minor valuations are known exactly, and no discreteness in the *order* of Γ is needed for it.

10.5 Three examples at the boundary

Example 10.7 (A positive matrix whose square root genuinely ramifies). Take

$$A = \begin{pmatrix} 1+t & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{C}((t))^{2 \times 2}, \quad (10.12)$$

positive definite because $x^*Ax = |x_1 + x_2|^2 + t|x_1|^2 > 0$ for $x \neq 0$. The minor minima are $\Delta_1 = 0$ and $\Delta_2 = v(t) = 1$, so the principal square root generates exactly $\mathbb{C}((t^{\frac{1}{2}}))$, a quadratic extension of $\mathbb{C}((t^{\mathbb{Z}}))$. Explicitly

$$B = \frac{A + \sqrt{t}I}{\sqrt{2+t+2\sqrt{t}}}, \quad (10.13)$$

whose denominator has a positive constant leading term and so introduces no exponent beyond $\frac{1}{2}\mathbb{Z}$; Cayley–Hamilton gives $B^2 = A$, and positivity of numerator and denominator gives $B \succeq 0$. Moreover $T = \text{tr}(B) = \sqrt{2+t+2\sqrt{t}}$ and $\sqrt{t} = (T^2 - 2 - t)/2$, which makes the primitive-trace phenomenon of [Section 11](#) directly visible.

Example 10.8 (A Gram witness can use a larger group than its product). With $t = t^\delta$, $u = t^\Omega$ in the rank-two group of [Example 9.2](#), put

$$C = \begin{pmatrix} t & 1 \\ 0 & u \end{pmatrix}, \quad A = C^*C = \begin{pmatrix} t^2 & t \\ t & 1+u^2 \end{pmatrix}.$$

Relative to $\Gamma = \mathbb{Z}\delta + \mathbb{Z}\Omega$ the minor valuations are $\Delta_1 = 0$ and $\Delta_2 = 2\delta + 2\Omega$, so the square root needs no extension while the fourth root generates one of degree 2. But the exact input support group of A alone is $H = \mathbb{Z}\delta + 2\mathbb{Z}\Omega$, and relative to H the value $2\delta + 2\Omega$ is not in $2H$, so $A^{1/2}$ generates a quadratic extension of K_H and recovers the missing scale u . This does not contradict [Corollary 9.19](#): the displayed Gram witness C already uses that scale. It is why every statement here names both the input matrix and the base group, and why [Proposition 2.3](#) is worth distinguishing from [Proposition 2.2](#).

Example 10.9 (The degree bound is attained in every rank). Let $\Gamma = \mathbb{Z}^r$ with the lexicographic order and standard basis $e_1, \dots, e_r$, each positive, and let $A = \text{diag}(t^{e_1}, \dots, t^{e_r})$. The nonzero eigenvalue valuations generate all of $\Gamma/q\Gamma$, and their partial sums, which are the determinantal values after ordering, generate the same subgroup. Hence $[K_\Gamma(A^{1/q}) : K_\Gamma] = q^r$. This rules out any degree bound depending only on q and not on the rank.

11 The trace is a primitive generator

The extension of [Theorem 10.4](#) is generated by the n^2 entries of $A^{1/q}$. It is generated by one of their linear functionals.

11.1 A positive-coset lemma

Lemma 11.1 (Positive radical sums retain every generator). *Let $a_1, \dots, a_r \in F_\Gamma$ be positive, let $q \geq 2$, and write $b_j = a_j^{1/q} = t^{\gamma_j/q} u_j$ as in (10.1). Put $\Lambda = \Gamma + \sum_j \mathbb{Z}\gamma_j/q$. Then for any strictly positive $w_j \in F_\Gamma$,*

$$K_\Gamma\left(\sum_{j=1}^r w_j b_j\right) = K_\Lambda. \quad (11.1)$$

Proof. Let $G = \Lambda/\Gamma$ and $h_j = \gamma_j/q + \Gamma$; these classes generate G . Choose a representative η_h for each class occurring among the h_j . The sum $T = \sum_j w_j b_j$ has the coset decomposition

$$T = \sum_h t^{\eta_h} s_h, \quad s_h = \sum_{j:h_j=h} w_j t^{\gamma_j/q - \eta_h} u_j \in F_\Gamma,$$

and every summand in s_h is positive in the ordered field F_Γ , so $s_h > 0$ and in particular $s_h \neq 0$ for each occurring class.

By Lemma 10.2 every automorphism of K_Λ/K_Γ is a character $\chi : G \rightarrow \mu_q$. If such an automorphism fixes T , then

$$0 = \chi(T) - T = \sum_h (\chi(h) - 1) t^{\eta_h} s_h.$$

Distinct cosets have disjoint supports, so every coefficient $\chi(h) - 1$ vanishes; since the occurring h_j generate G , the character is trivial. Thus T has trivial stabilizer in the Galois group, and the fundamental correspondence gives $[K_\Gamma(T) : K_\Gamma] = |G|$, which is the asserted equality. $\square$

Theorem 11.2 (Primitive trace). *Under the hypotheses of Theorem 10.4,*

$$K_\Gamma(\text{tr}(A^{1/q})) = K_\Gamma(A^{1/q}) = K_{\Gamma_{A,q}}. \quad (11.2)$$

More generally, if $\lambda_1, \dots, \lambda_r$ are the positive eigenvalues of A , then every sum $\sum_j a_j \lambda_j^{1/q}$ with $a_j \in F_\Gamma$ and $a_j > 0$ is a primitive generator of the same field.

Proof. For the principal root (10.5), trace invariance gives $\text{tr}(B) = \sum_{j=1}^r \lambda_j^{1/q}$. Apply Lemma 11.1 with $w_j = 1$, and use Theorem 10.3 together with (10.10). The weighted statement is the same lemma with arbitrary strictly positive w_j . $\square$

The proof identifies a mechanism rather than invoking an abstract primitive-element theorem. In a general finite extension most linear combinations may be primitive while special choices fail; here the *specific* unweighted trace always succeeds, because positivity makes its coset support a generating set. Strictly positive coefficients are the whole content of the hypothesis, as Warning 11.5 shows.

Corollary 11.3 (The corresponding real field). *With $T = \text{tr}(A^{1/q})$ and $\Lambda = \Gamma_{A,q}$, we have $F_\Gamma(T) = F_\Lambda$.*

Proof. Both fields are ordered and $T \in F_\Lambda$. Adjoining i has degree 2 over each of these real fields, so the tower formula applied to $K_\Gamma(T) = K_\Lambda$ gives $[F_\Gamma(T) : F_\Gamma] = [K_\Lambda : K_\Gamma] = |\Lambda/\Gamma|$, which is $[F_\Lambda : F_\Gamma]$ by Lemma 10.2. $\square$

11.2 An explicit irreducible orbit polynomial

Let $b_j = \lambda_j^{1/q}$, $G = \Gamma_{A,q}/\Gamma$ and $h_j = v(b_j) + \Gamma$. The minimal polynomial of the trace over K_Γ is

$$\mathcal{P}_{A,q}(X) = \prod_{\chi \in \text{Hom}(G, \mu_q)} \left(X - \sum_{j=1}^r \chi(h_j) b_j \right). \quad (11.3)$$

Its coefficients lie in K_Γ because the Galois group permutes the factors, and the factors are distinct by [Lemma 11.1](#), so the polynomial is irreducible of degree $|R_q(A)|$. It is a formula inside a finite extension, not necessarily a fast coefficient-extraction algorithm.

Example 11.4 (A biquadratic field encoded in one positive trace). Let $\Gamma = \mathbb{Z}\delta \oplus \mathbb{Z}\Omega$ with $0 < n\delta < \Omega$ for every positive integer n , and let $A = \text{diag}(t^\delta, t^\Omega)$ with $q = 2$. The minor valuations are $\Delta_1 = \delta$ and $\Delta_2 = \delta + \Omega$, and their classes span a subgroup of order 4 in $\Gamma/2\Gamma$. So $B = A^{1/2}$ generates a biquadratic extension, and so does $T = t^{\delta/2} + t^{\Omega/2}$. Writing $t = t^\delta$ and $u = t^\Omega$, the minimal polynomial is

$$X^4 - 2(t+u)X^2 + (t-u)^2, \quad (11.4)$$

whose degree 4 is certified by the two independent exponent classes and not by an unproved irreducibility guess about the displayed polynomial.

Warning 11.5 (Positivity cannot be dropped). For $A = t^\delta I_2$ with $\delta \notin 2\Gamma$, the matrix $C = \text{diag}(t^{\delta/2}, -t^{\delta/2})$ is a Hermitian square root generating a quadratic extension, and $\text{tr}(C) = 0$. It is not the positive root. [Theorem 11.2](#) concerns the principal positive root and strictly positive weights, never an arbitrary branch or a signed sum.

11.3 Several positive roots at once

Corollary 11.6 (Joint ramification and a joint primitive element). *Let $A_1, \dots, A_m$ be positive semidefinite Hermitian matrices over K_Γ , of possibly different sizes, and let $q_\ell \geq 2$. Put*

$$\Lambda = \Gamma + \sum_{\ell=1}^m \sum_{k=1}^{\text{rank}(A_\ell)} \mathbb{Z} \frac{\Delta_k(A_\ell)}{q_\ell}.$$

Then the field generated by all entries of all A_ℓ^{1/q_ℓ} is K_Λ , of degree $|\Lambda/\Gamma|$, and the single scalar $T = \sum_\ell \text{tr}(A_\ell^{1/q_\ell})$ is a primitive generator. No commutation assumption on the A_ℓ is needed.

Proof. The compositum of the individual monomial extensions contains the monomials generating Λ , and the quotient is finite and killed by $Q = \text{lcm}(q_1, \dots, q_m)$, so [Lemma 10.2](#) identifies the compositum with K_Λ . Decompose the finitely many positive scalar spectral roots occurring in the traces into cosets of Γ ; every occurring coefficient sum is positive and the occurring classes generate Λ/Γ , so the character argument of [Lemma 11.1](#), now with μ_Q , proves primitivity. $\square$

11.4 Back inside the surcomplex numbers

Corollary 11.7 (Exact-support surcomplex spectral calculus). *Let A be a finite surcomplex matrix and let H be its exact input support group, as in [Proposition 2.3](#). If A is Hermitian, it has real surreal eigenvalues and a unitary eigenbasis whose normal-form exponents stay in H ; the same support assertion holds for normal diagonalization and for the SVD of an arbitrary finite surcomplex matrix. If A is in addition positive semidefinite, then for every $q \geq 2$*

$$K_H(A^{1/q}) = K_{H+\sum_k \mathbb{Z}\Delta_k(A)/q} = K_H(\text{tr}(A^{1/q})),$$

and the degree over K_H is the order of the subgroup generated by the classes $\Delta_k(A) + qH$ in H/qH .

Proof. Apply [Theorems 9.12, 9.18, 10.4 and 11.2](#) and [Corollary 9.15](#) inside the set-sized field K_H , then transport by the normal-form embedding of [Proposition 2.3](#). The positive branch agrees with the ambient surreal positive root by ordered-field uniqueness together with the uniqueness proof in [Theorem 10.3](#). $\square$

The ambient surreal exponent group is divisible, so the full class field has room for all of these fractional scales; the content of the corollary is not that positive roots fail to exist in $\mathbf{No}[i]$. It is their exact *relative cost* over the smallest Hahn workspace containing the data — a cost that becomes invisible the moment one replaces H by its divisible hull, which is precisely what [Proposition 2.2](#) does.

12 Rank at a chosen scale

12.1 Exact rank and standard-part rank

Corollary 12.1 (Rank after reduction). *If $A \in \mathcal{O}^{m \times n}$ has nonzero singular scales $\gamma_1, \dots, \gamma_r$, then every $\gamma_j \geq 0$ and*

$$\text{rank}_{\mathbb{C}}(\text{st } A) = \#\{j : \gamma_j = 0\}.$$

Proof. By [Lemma 7.2](#), $v(\sigma_1) = v(A) \geq 0$, so all singular values are integral. Reduce an SVD. Its unitary factors reduce to invertible ordinary unitary matrices, and its diagonal entries reduce to nonzero numbers exactly at valuation zero. $\square$

An invertible matrix can therefore have a singular standard part. Conversely, an invertible standard part certifies invertibility of an integral square matrix. The inverse need not be integral in the first situation, but is integral in the second: the latter is exactly the case in which all its singular scales are zero.

12.2 An intrinsic input filtration

Fix $\theta \in \Gamma$ and define an $\mathcal{O}$ -submodule of the input lattice by

$$M_{\theta}(A) = \{x \in \mathcal{O}^n : Ax \in t^{\theta} \mathcal{O}^m\}. \quad (12.1)$$

Its image $\text{st}(M_{\theta}(A))$ under coordinatewise reduction is a $\mathbb{C}$ -linear subspace of $\mathbb{C}^n$. Also define

$$M_{>\theta}(A) = \{x \in \mathcal{O}^n : v(Ax) > \theta\}.$$

These definitions work even when A or t^{θ} is nonintegral; only the input vectors are required to lie in the integral lattice.

Theorem 12.2 (Filtration dimensions). *For every $\theta \in \Gamma$,*

$$\dim_{\mathbb{C}} \text{st}(M_{\theta}(A)) = n - \#\{j : \gamma_j < \theta\}, \quad (12.2)$$

$$\dim_{\mathbb{C}} \text{st}(M_{>\theta}(A)) = n - \#\{j : \gamma_j \leq \theta\}. \quad (12.3)$$

Only the nonzero singular values enter the counts.

Proof. Use an SVD $A = U\Sigma V^*$. Both unitary factors preserve integral lattices and their reductions are invertible. Write $x = Vy$. For $j \leq r$, condition (12.1) becomes

$$v(y_j) \geq \max(0, \theta - \gamma_j).$$

The kernel coordinates $j > r$ are unrestricted in $\mathcal{O}$. The reduction of y_j must be zero exactly when $\gamma_j < \theta$; all other reduced coordinates can be chosen arbitrarily. This gives (12.2). For the strict condition, reduction of y_j vanishes exactly when $\gamma_j \leq \theta$. Multiplication by the invertible $\text{st } V$ preserves the dimension, proving (12.3). $\square$

The two formulas deliberately differ at the boundary. A direction of singular scale exactly θ remains visible when the output threshold is strict, and disappears if only outputs of strictly larger magnitude are counted. It is helpful to fix explicit notation:

$$r_{<\theta}(A) = \#\{j : \gamma_j < \theta\}, \quad r_{\leq\theta}(A) = \#\{j : \gamma_j \leq \theta\}.$$

These are not replacements for algebraic rank. They describe how much rank is visible at a specified valuation resolution. In the integral case, $r_{\leq 0}(A) = \text{rank}(\text{st } A)$.

Example 12.3 (Three genuinely different levels). Choose a workspace containing 1 and ω in its value group and let

$$A = \text{diag}(1, t^2, t^\omega), \quad t = \omega^{-1}.$$

Its exact rank is three and its standard-part rank is one. Its closed-scale rank $r_{\leq\theta}$ is one for $0 \leq \theta < 2$, two for $2 \leq \theta < \omega$, and three for $\theta \geq \omega$. No finite integer cutoff N reaches the last direction. This is not a statement about a slow numerical convergence rate; it is an ordered-group fact, $N < \omega$ for every ordinary N .

13 A root-free scale algorithm and its computational contract

13.1 Minimum-valuation elimination

The determinantal formula is finite, but enumerating all minors is expensive. A valuation-aware elimination computes the same scale tuple without singular vectors or algebraic root extraction.

Proposition 13.1 (Exact scale elimination). *Suppose exact field arithmetic, exact zero testing, and comparison of valuations of nonzero elements are available. The following algorithm terminates after at most $\min(m, n)$ pivots and returns $\gamma_1, \dots, \gamma_r$ in nondecreasing order.*

In the active submatrix, choose a nonzero entry of minimum valuation and move it to the leading position by row and column swaps. Clear the remaining entries in its column by row operations, and then clear its row by column operations. Record its valuation and recurse on the trailing submatrix. Stop when the active submatrix is zero.

Proof. Write one active block as

$$\begin{pmatrix} a & b \\ c & D \end{pmatrix}, \quad v(a) = \alpha \leq v(\text{every entry}).$$

The row multipliers c_i/a and column multipliers b_j/a are integral. Their elementary matrices and inverses therefore lie in $\text{GL}(\mathcal{O})$. The two clearings leave the Schur complement

$$D - ca^{-1}b. \tag{13.1}$$

Every term in it has valuation at least α , because $v(c_i) + v(b_j) - v(a) \geq \alpha$. Cancellation can raise the valuation, but cannot lower it. Thus subsequent pivot valuations are nondecreasing. After finitely many steps the result is a diagonal matrix and a zero block. All transformations were integral and invertible, so they preserve the Δ_k . The diagonal pivot valuations must therefore be their successive differences, which are the singular scales by Theorem 7.5. $\square$

The algorithm uses $O(mn \min(m, n))$ field operations, apart from pivot comparisons. This counts algebraic operations; it is not a bit-complexity bound for arbitrary Hahn-series representations. A single exact zero test may already be unavailable for a given symbolic representation. Moreover, the elimination matrices are generally not unitary: this algorithm returns the singular *scales*, not the singular values or the best low-rank approximation itself.

13.2 What must be provided by a symbolic scalar representation

The algorithm has four logically separate requirements: a representation closed under the field operations it performs, exact zero testing, exact leading-exponent access for nonzero outputs, and an effective order on those exponents. Rational functions over a finite rational or Gaussian-rational monomial field can meet these requirements. A name for an arbitrary surreal number does not automatically meet any of them. This is the same kind of denotation-versus-computation distinction emphasized by the repository’s computer-algebra strand [1].

Cancellation is the principal practical issue. A representation that stores only the current leading term is insufficient: (13.1) may cancel it. Correct implementations must either compute further exact coefficients, attach a certified remainder bound, or report that the pivot scale is unresolved. Reporting a zero pivot merely because all currently stored coefficients vanish is unsound.

13.3 Two further contracts: spectral descent and ramification

The support-certified spectral construction of Section 9 is more demanding than the scale algorithm. It needs exact arithmetic and conjugation, exact zero tests, leading exponents and leading coefficients for nonzero expressions, ordinary constant Hermitian diagonalization, and a representation of strongly summable substitutions together with their support certificates. An exact representation need not provide any of these effectively merely because it can denote a surreal number. Example 9.14 is the concrete obstruction: a procedure that inspects more and more coefficients never reaches the first nonscalar scale, whereas the exact trace subtraction (9.10) reaches it in one step. A practical implementation should also keep two outputs apart. A matrix identity such as $U^*AU = D$ is an *algebraic* certificate; a support certificate is what licenses the Hahn expressions occurring in it. Finite truncation can test an identity to a declared order, but cannot certify exact zero, all coefficients, or well-ordering of an arbitrary support.

The determinantal ramification computation of Section 10 needs much less: exact valuations of finitely many minors, and a way to compute the finite subgroup generated by specified classes modulo $q\Gamma$. In a finite integer lattice (10.11) makes this a Smith normal form, hence a concrete algorithm.

Warning 13.2 (Membership in $q\Gamma$ is extra data). For an abstract ordered group Γ presented only through a black-box comparison of elements, membership of Δ_k in $q\Gamma$ is additional group-theoretic data and need not be decidable. Order comparison alone does not answer it. Consequently (10.9) — “ $A^{1/q}$ is defined over K_Γ if and only if every $\Delta_k(A)$ lies in $q\Gamma$ ” — is an exact mathematical criterion that a given representation may nevertheless be unable to evaluate, even when it can compute every Δ_k exactly.

This is the same denotation-versus-computation distinction as above, one tier further out, and it is exactly the question that the repository’s computer-algebra report formalizes as a hierarchy of representation capabilities [4]. That report’s own inventory already makes the matching distinction on the implementation side: a ramified root is representable in a model whose exponent lattice is $\mathbb{Q}^r$ and is not representable in one whose exponent lattice is a finite-rank free abelian group — which is the non-divisibility distinction of Section 10 seen from the other end. Equation (10.9) is therefore a useful concrete instance of a decision that needs a capability

tier the black-box order interface does not supply. Nothing here claims to solve that report's representation or computability problems.

13.4 Why finite truncation cannot certify every exact rank

Fix any precision threshold θ . Choose $\eta > \theta$. The matrices

$$\text{diag}(1, 0), \quad \text{diag}(1, t^\eta)$$

agree at all exponents at most θ but have different exact ranks. Thus finite-precision coefficient access alone cannot decide exact rank uniformly. An exact representation may still decide it, for example by computing a nonzero symbolic determinant, but truncation alone cannot.

The right numerical-style output is therefore often a certified rank profile up to a declared cutoff, with the remaining directions unresolved. The perturbation theorem in [Section 14](#) specifies exactly what precision is sufficient for such a certificate. A finite exact rational kernel for testing the elimination and Gram identities is included in the archive; its scope is described in [Section B](#).

14 Condition numbers and perturbation certificates

14.1 Distance to singularity is a minimum, not merely an infimum

This subsection again holds over any real closed F and $K = F(i)$.

Theorem 14.1 (Distance to singularity). *For an invertible square A ,*

$$\|A^{-1}\|_{\text{op}} = \sigma_n(A)^{-1}, \quad \min_{A+E \text{ singular}} \|E\|_{\text{op}} = \sigma_n(A). \quad (14.1)$$

The latter minimum is attained by $E = -\sigma_n u_n v_n^$ for any matched least-singular vector pair.*

Proof. Invert the SVD to obtain the first formula. The stated perturbation deletes one diagonal entry of Σ and has norm σ_n . Conversely, if $A + E$ is singular, choose a unit vector x in its kernel. Then

$$\|E\|_{\text{op}} \geq \|Ex\| = \|Ax\| \geq \sigma_n.$$

□

Define the ordered spectral condition number by

$$\kappa(A) = \|A\|_{\text{op}} \|A^{-1}\|_{\text{op}} = \frac{\sigma_1}{\sigma_n} \geq 1.$$

It is an element of F , not necessarily an ordinary real. In a Hahn workspace,

$$v(\kappa(A)) = \gamma_1 - \gamma_n. \quad (14.2)$$

Consequently $\kappa(A)$ is infinite precisely when $\gamma_1 < \gamma_n$. Matrices whose nonzero singular values all have the same valuation have finite condition number, although its ordinary leading coefficient can still be large. Overall scale and relative conditioning are separate: $t^{-\omega}I$ has infinite norm but condition number one.

Theorem 14.2 (Inverse and linear-system perturbation). *Let A be invertible and put $q = \|A^{-1}\|_{\text{op}} \|E\|_{\text{op}}$. If $q < 1$, then $A + E$ is invertible,*

$$\|(A + E)^{-1}\|_{\text{op}} \leq \frac{\|A^{-1}\|_{\text{op}}}{1 - q}, \quad (14.3)$$

$$\|(A + E)^{-1} - A^{-1}\|_{\text{op}} \leq \frac{\|A^{-1}\|_{\text{op}}^2 \|E\|_{\text{op}}}{1 - q}. \quad (14.4)$$

If $Ax = b \neq 0$ and $(A + E)(x + \delta x) = b + f$, then

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A)}{1 - q} \left(\frac{\|f\|}{\|b\|} + \frac{\|E\|_{\text{op}}}{\|A\|_{\text{op}}} \right). \quad (14.5)$$

Proof. For every vector x ,

$$\|(A + E)x\| \geq (\sigma_n(A) - \|E\|_{\text{op}}) \|x\|.$$

The coefficient is positive under the hypothesis. The map is injective, hence invertible in finite dimension, and this inequality gives (14.3). The finite resolvent identity

$$(A + E)^{-1} - A^{-1} = -(A + E)^{-1}EA^{-1}$$

gives (14.4). Also $\delta x = (A + E)^{-1}(f - Ex)$. Apply (14.3), divide by $\|x\|$, and use $\|b\| \leq \|A\|_{\text{op}} \|x\|$ to obtain (14.5). $\square$

No geometric series was used. This matters because the ordered inequality $q < 1$ is not by itself a Hahn-summability hypothesis.

14.2 A certified precision theorem for singular scales

Return to a Hahn workspace. When $v(A) = \eta$, every entry of A has valuation at least η ; this is the matrix precision convention below.

Theorem 14.3 (Preserving visible singular data). *Let $A, B = A + E$ have the same dimensions. If $v(E) > \gamma_k(A)$ for a nonzero singular value $\sigma_k(A)$, then for all $1 \leq j \leq k$,*

$$v(\sigma_j(B)) = \gamma_j(A), \quad \text{lc}(\sigma_j(B)) = \text{lc}(\sigma_j(A)).$$

More generally, if $v(E) > \theta$, then

$$r_{\leq \theta}(A) = r_{\leq \theta}(B), \quad r_{< \theta}(A) = r_{< \theta}(B).$$

The exact ranks need not be equal.

Proof. By Lemma 7.2, $v(\|E\|_{\text{op}}) = v(E)$ for $E \neq 0$. The ordered singular-value estimate (5.8) therefore implies

$$v(\sigma_j(B) - \sigma_j(A)) \geq v(E).$$

An error of strictly larger valuation cannot alter a nonzero leading term. This proves the first assertion. If a singular value has valuation greater than θ , or is zero, adding an error of valuation greater than θ keeps it above that threshold. Singular values at or below the threshold retain their leading terms. Apply this coordinatewise to the ordered singular lists for both asserted counts. The case $E = 0$ is immediate. $\square$

Strictness is essential. For $A = \text{diag}(1, t^\theta)$ with $\theta > 0$, a perturbation $E = \text{diag}(0, -t^\theta)$ has exactly valuation θ and destroys the second singular direction. Even under the strict hypothesis, a zero singular value can become a nonzero value of a still smaller magnitude; $\text{diag}(1, 0)$ and $\text{diag}(1, t^\eta)$ with $\eta > \theta$ show why exact rank is not certified. The theorem certifies the rank *visible at the stated scale*.

15 Nonnormal matrices and ordered pseudospectra

Eigenvalues alone do not control a general matrix. For example,

$$N_L = \begin{pmatrix} 0 & L \\ 0 & 0 \end{pmatrix}$$

has only the eigenvalue zero, while $\|N_L\|_{\text{op}} = |L|$ can be infinite. The distinction between eigenvalues and singular values is as important in the surcomplex field as it is over $\mathbb{C}$.

For a square matrix and an ordered tolerance $\varepsilon \in F_{\geq 0}$, define the closed ordered pseudospectrum by

$$\Lambda_\varepsilon(A) = \{z \in K : \sigma_n(zI - A) \leq \varepsilon\}. \quad (15.1)$$

Proposition 15.1 (Exact perturbation interpretation). *A scalar z belongs to $\Lambda_\varepsilon(A)$ if and only if there is E with $\|E\|_{\text{op}} \leq \varepsilon$ for which z is an eigenvalue of $A + E$. If $z \notin \text{spec } A$ and $\varepsilon > 0$, this is equivalent to*

$$\|(zI - A)^{-1}\|_{\text{op}} \geq \varepsilon^{-1}.$$

For normal A ,

$$\Lambda_\varepsilon(A) = \bigcup_{\lambda \in \text{spec } A} \{z : |z - \lambda| \leq \varepsilon\}. \quad (15.2)$$

Proof. Apply the distance-to-singularity argument to $zI - A$. If it is already singular, use $E = 0$. Otherwise [Theorem 14.1](#) gives an attained perturbation of norm $\sigma_n(zI - A)$ making it singular; changing its sign converts it to a perturbation of A . The lower bound follows by applying $zI - A$ to an eigenvector of $A + E$. The inverse characterization is [\(14.1\)](#). For normal A , unitary diagonalization gives singular values of $zI - A$ equal to the numbers $|z - \lambda_j|$, proving [\(15.2\)](#). $\square$

Example 15.2 (An infinitesimal perturbation with a larger spectral effect). Let

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 \\ t^2 & 0 \end{pmatrix}.$$

Then $\|E\|_{\text{op}} = t^2$, whereas $N + E$ has eigenvalues $\pm t$. Thus $t \in \Lambda_{t^2}(N)$ although its distance from the unperturbed spectrum is $t > t^2$. More generally, for nonzero infinitesimal z , $zI - N$ has $\Delta_1 = 0$ and $\Delta_2 = 2v(z)$, so its least singular value has valuation $2v(z)$. The nonlinear sensitivity is visible directly in the determinantal scale profile of the resolvent matrix.

Here ε is an actual ordered magnitude, not just a value-group element. A valuation threshold gives a coarser notion. In particular, replacing the ordered disks in [\(15.2\)](#) by valuation balls would change the assertion. No fine-continuous contour is needed in the pseudospectral argument.

16 Spectral subspaces and infinitesimal gaps

Individual eigenvectors are not canonical, especially at multiplicities. Orthogonal spectral projectors are the better perturbation objects. The following finite-dimensional Frobenius version of a Davis–Kahan bound is proved directly over F ; the classical context is [\[11, 12\]](#). No inverse trigonometric function is needed to measure the subspace displacement.

Theorem 16.1 (A separated-subspace bound). *Let $H = H^*$, $B = H + E = B^*$, and let P be an orthogonal projector onto an H -invariant subspace. Let Q project onto a B -invariant subspace of rank k . Suppose that every eigenvalue λ of $H|_{\text{im}(I-P)}$ and every eigenvalue μ of $B|_{\text{im } Q}$ satisfy*

$$|\lambda - \mu| \geq \delta > 0.$$

Then

$$\|(I - P)Q\|_F \leq \frac{\|E\|_F}{\delta}. \quad (16.1)$$

If $\text{rank } P = \text{rank } Q$, also

$$\|P - Q\|_F \leq \frac{\sqrt{2}\|E\|_F}{\delta}. \quad (16.2)$$

Proof. Choose orthonormal eigenbasis matrices Y for $\text{im}(I - P)$ and X for $\text{im } Q$, so $HY = Y\Lambda$ and $BX = XM$, with real diagonal Λ, M . Put $C = Y^*X$. Multiplying $HX + EX = XM$ on the left by Y^* gives the Sylvester equation

$$\Lambda C - CM = -Y^*EX.$$

Entrywise, $(\lambda_a - \mu_b)c_{ab} = -(Y^*EX)_{ab}$, hence

$$\|C\|_F^2 \leq \delta^{-2} \|Y^*EX\|_F^2 \leq \delta^{-2} \|E\|_F^2.$$

The last inequality follows because X and Y have orthonormal columns; extend them to unitary matrices and observe that taking a submatrix cannot increase the sum of squared moduli. Since $(I - P)Q = YCX^*$, $\|(I - P)Q\|_F = \|C\|_F$. This proves (16.1).

For equal ranks,

$$\|P - Q\|_F^2 = \text{tr } P + \text{tr } Q - 2\text{tr}(PQ) = 2\|(I - P)Q\|_F^2,$$

which gives (16.2). $\square$

Corollary 16.2 (Using an unperturbed cluster gap). *Select a nonempty proper set of eigenvalue indices of H , separated from all complementary indices by at least $g > 0$. Let P project onto this cluster, and let Q select the same indices in the decreasing eigenvalue list of $H + E$. If $e = \|E\|_{\text{op}} < g/2$, then*

$$\|P - Q\|_F \leq \frac{\sqrt{2}\|E\|_F}{g - e}.$$

Proof. The eigenvalue estimate in Theorem 5.2 moves each selected eigenvalue by at most e . Its distance from an unselected eigenvalue of H is at least $g - e$. No eigenvalue can cross the external gap under $e < g/2$. The chosen subspaces have the same dimension, so apply Theorem 16.1 with $\delta = g - e$. $\square$

The Frobenius separation condition is all that was used in the proof. We are not asserting every operator-norm variant of the classical Davis–Kahan theorem under an arbitrary separation hypothesis. The Frobenius estimate is sufficient for an important surreal consequence: if $v(E) > v(g)$ in a Hahn workspace, then the projector displacement is infinitesimal. Indeed, $\|E\|_F/g$ then has positive valuation, so $e < g/2$ and the displayed bound has positive valuation.

Example 16.3 (Infinitesimal error, ordinary subspace rotation). Consider

$$H = \begin{pmatrix} 0 & 0 \\ 0 & t^2 \end{pmatrix}, \quad B = \frac{t^2}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

They have the same eigenvalues $0, t^2$. Their zero-eigenspace projectors are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Exact multiplication gives

$$\|B - H\|_{\text{op}} = \frac{t^2}{\sqrt{2}}, \quad \|P - Q\|_{\text{F}} = 1, \quad \|P - Q\|_{\text{op}} = \frac{1}{\sqrt{2}}.$$

Thus the matrix error is infinitesimal but the subspace change is not. Its size is comparable to the gap t^2 , so the gap-relative smallness hypothesis fails. Discarding the common infinitesimal factor before asking about eigenvectors would conceal the entire phenomenon.

16.1 Singular subspaces and a warning about Gram squaring

Right singular subspaces are eigenspaces of A^*A . Under $A \mapsto A + E$,

$$(A + E)^*(A + E) - A^*A = A^*E + E^*A + E^*E,$$

so its operator norm is at most $2\|A\|_{\text{op}}\|E\|_{\text{op}} + \|E\|_{\text{op}}^2$. One can therefore apply [Theorem 16.1](#) to squared-singular-value gaps. This is valid but can be quantitatively wasteful: it changes gaps $\sigma_i - \sigma_j$ into $\sigma_i^2 - \sigma_j^2$. A direct singular-subspace or Hermitian-dilation argument can yield better bounds, but no sharp Wedin theorem is claimed here. The distinction is particularly relevant when both singular values are infinitesimal.

17 Degenerate clusters and Schur-complement corrections

A cluster of repeated or nearly repeated eigenvalues cannot generally be understood by retaining only the perturbation inside that cluster. Coupling to its complement can create a smaller, second-order splitting. The right starting point is an exact finite identity.

Proposition 17.1 (Exact effective matrix). *Suppose a Hermitian matrix is written in an orthogonal decomposition as*

$$H = \begin{pmatrix} \lambda_0 I + E_{11} & E_{12} \\ E_{21} & D + E_{22} \end{pmatrix}, \quad E_{21} = E_{12}^*.$$

For $z \in K$ such that $D + E_{22} - zI$ is invertible,

$$\det(H - zI) = \det(D + E_{22} - zI) \det S(z), \tag{17.1}$$

$$S(z) = (\lambda_0 - z)I + E_{11} - E_{12}(D + E_{22} - zI)^{-1}E_{21}. \tag{17.2}$$

If the ordered distance from z to every eigenvalue of $D + E_{22}$ is at least $\delta > 0$, then

$$\|E_{12}(D + E_{22} - zI)^{-1}E_{21}\|_{\text{op}} \leq \frac{\|E_{12}\|_{\text{op}}^2}{\delta}. \tag{17.3}$$

Proof. Multiply $H - zI$ on the left by the determinant-one block matrix

$$\begin{pmatrix} I & -E_{12}(D + E_{22} - zI)^{-1} \\ 0 & I \end{pmatrix}.$$

The resulting matrix is block lower triangular with diagonal blocks $S(z)$ and $D + E_{22} - zI$, proving (17.1). Diagonalization of the Hermitian complement shows that its shifted inverse has norm at most δ^{-1} . Apply the product norm bounds and $\|E_{21}\|_{\text{op}} = \|E_{12}\|_{\text{op}}$. $\square$

For real z outside the complementary spectrum, $S(z)$ is Hermitian. It is an eigenvalue-dependent effective matrix, not in general one fixed matrix whose eigenvalues already solve the original problem. Freezing z or dropping the inverse correction requires a separate error estimate. Formula (17.3) supplies one such estimate: the correction has the scale of the square of the coupling divided by the complementary gap. If that gap is itself infinitesimal, a seemingly second-order term can be as important as a first-order term.

Example 17.2 (A spectral shift invisible in the small diagonal block). Let

$$H_t = \begin{pmatrix} 0 & t \\ t & 1 \end{pmatrix}.$$

Its eigenvalues and low-eigenvalue projector are exactly

$$\lambda_{\pm} = \frac{1 \pm \sqrt{1 + 4t^2}}{2}, \quad (17.4)$$

$$P_- = \frac{\lambda_+ I - H_t}{\lambda_+ - \lambda_-}. \quad (17.5)$$

The small eigenvalue is

$$\lambda_- = -t^2 + t^4 - 2t^6 + 5t^8 + O_v(t^{10}). \quad (17.6)$$

Here $O_v(t^{10})$ denotes a Hahn remainder of valuation at least 10; t is not a real variable tending to zero. The expansion follows either from the binomial series justified in Section 18, or by substituting the displayed finite polynomial into $\lambda^2 - \lambda - t^2$ and checking the remainder order, then selecting the root with positive valuation.

The direct perturbation inside the first coordinate is zero. Nevertheless, (17.2) gives the scalar equation $-z - t^2/(1 - z) = 0$, whose leading solution is $z = -t^2$. An eigenvector may be taken as $(1, \lambda_-/t)^T$, displaying an order- t mixing with a spectral shift of order t^2 .

The companion matrix

$$\tilde{H}_t = \begin{pmatrix} t^2 & t \\ t & 1 \end{pmatrix} = \begin{pmatrix} t \\ 1 \end{pmatrix} \begin{pmatrix} t & 1 \end{pmatrix}$$

is positive semidefinite of exact rank one. In its effective scalar at $z = 0$, the direct t^2 term and the coupling correction t^2 cancel exactly. The difference between a small eigenvalue and a zero eigenvalue can therefore depend on cancellation between terms of the same scale. Neither leading entry valuations nor a generic “second-order is negligible” rule resolves it.

18 Finite functional calculus versus infinite Hahn series

18.1 Finite spectral calculus needs no convergence

Let H be Hermitian with distinct eigenvalues in a finite set S . For any prescribed scalars $f(\lambda) \in K$, define

$$f(H) = \sum_{\lambda \in S} f(\lambda) P_{\lambda}. \quad (18.1)$$

Lagrange interpolation gives a polynomial $p \in K[X]$ of degree less than $|S|$ with $p(\lambda) = f(\lambda)$; thus $f(H) = p(H)$. The construction preserves finite sums and products and depends only on the values of f at the spectrum. For real-valued spectral data it gives a Hermitian matrix; for nonnegative data it gives a positive semidefinite matrix. It also yields

$$\|f(H)\|_{\text{op}} = \max_{\lambda \in S} |f(\lambda)|.$$

For $z \notin S$, the resolvent is

$$(zI - H)^{-1} = \sum_{\lambda \in S} \frac{P_\lambda}{z - \lambda}.$$

All of this is finite algebra. It does not assert that an arbitrary scalar function already has a chosen surreal extension. In particular, one cannot use (18.1) to define unspecified values such as a global surreal sine at an infinite argument by circular reasoning. One must first supply the values to be interpolated.

18.2 A genuine matrix Hahn-summability theorem

A family of Hahn series is *strongly summable* if the union of its supports is well ordered and every exponent occurs in only finitely many members (the notion of Definition 9.1). Its sum is then defined coefficientwise by finite sums. For matrices this means entrywise strong summability, equivalently the same condition on the finitely many entry families together.

We use the standard support lemma underlying Hahn-field arithmetic: if $S \subseteq \Gamma_{>0}$ is well ordered, its additive monoid

$$S^* = \{0\} \cup \{s_1 + \cdots + s_k : k \geq 1, s_j \in S\}$$

is well ordered, and every exponent has only finitely many representations as an ordered finite sum of members of S . This is the positive-support form of the Neumann lemma; see the support development in Gonshor, Chapter 5A [5], and Lemma 9.4 for a proof from the ordered-word lemma. The lemma does not assert that $k\rho$ is cofinal in Γ for a positive ρ .

Theorem 18.1 (Matrix inverse and square-root series). *If $E \in K_\Gamma^{n \times n}$ has $v(E) > 0$, then*

$$\sum_{k \geq 0} E^k \tag{18.2}$$

is strongly Hahn-summable and equals $(I - E)^{-1}$. If in addition $E = E^$, then*

$$\sum_{k \geq 0} \binom{1/2}{k} E^k \tag{18.3}$$

is strongly Hahn-summable and equals the positive square root of $I + E$.

Proof. The finite union of the entry supports of E is a well-ordered subset $S \subseteq \Gamma_{>0}$. An entry of E^k is a finite sum of products of k entries, so its support is contained in sums of k elements of S . The support lemma gives a well-ordered union and finitely many contributing lengths and tuples at every exponent. For a fixed length there are only finitely many matrix-index paths. This proves strong summability of both series, and justifies the coefficientwise rearrangements below.

The finite formal identity $(1 - X) \sum_{k \geq 0} X^k = 1$ is valid coefficientwise after substituting E , giving a two-sided inverse. Likewise the formal binomial identity makes the square of (18.3) equal to $I + E$. Its sum R is Hermitian, because every power of E is, and $v(R - I) > 0$. The norm of $R - I$ is therefore infinitesimal by Lemma 7.2; all eigenvalues of R are positive by Theorem 5.2. Uniqueness of the positive square root from Proposition 4.3 identifies R with $(I + E)^{1/2}$. $\square$

18.3 Two failures of a familiar shortcut

The ordered condition $\|E\|_{\text{op}} < 1$ does not imply strong Hahn-summability of (18.2). For $E = \frac{1}{2}I$, every nonzero entry of every power has exponent zero, so infinitely many members meet the same exponent. The Hahn summability criterion fails, although $(I - E)^{-1} = 2I$ exists algebraically. Ordinary analytic summation of real coefficients is a different operation.

Even when the Hahn series exists, it need not be a topological limit of its partial sums in the full surreal field. With $E = tI$ and $t = \omega^{-1}$,

$$(I - tI)^{-1} - \sum_{k=0}^N t^k I = \frac{t^{N+1}}{1 - t} I.$$

For every ordinary N , this error is larger in modulus than t^ω . Thus the partial sums fail the fine-surreal convergence test at the single tolerance t^ω .

There is no contradiction with internal valuation convergence in $K_{\mathbb{Q}} = \mathbb{C}((t^{\mathbb{Q}}))$: the natural-number exponents are cofinal in $\mathbb{Q}$, so these same partial sums do converge in that field's own valuation topology. In the larger workspace with value group $\mathbb{Q} + \mathbb{Q}\omega$, they do not, because the threshold ω is now present. Algebraic identities, strong summability, and workspace-relative topology are three different claims; only the first two were needed for [Theorem 18.1](#).

19 Two applications: regularization and finite tensors

19.1 Regularization as an explicit scale filter

For $\alpha > 0$, consider the regularized least-squares objective

$$J_\alpha(x) = \|Ax - b\|^2 + \alpha^2 \|x\|^2.$$

The matrix $A^*A + \alpha^2 I$ is positive definite, regardless of the rank of A . Define

$$x_\alpha = (A^*A + \alpha^2 I)^{-1} A^*b. \quad (19.1)$$

Expanding and using its normal equation yields

$$J_\alpha(x) = J_\alpha(x_\alpha) + (x - x_\alpha)^*(A^*A + \alpha^2 I)(x - x_\alpha).$$

Hence (19.1) is the unique minimizer over the whole field, with no compactness or coercivity argument required.

In singular coordinates,

$$x_\alpha = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \alpha^2} (u_j^* b) v_j, \quad Ax_\alpha = \sum_{j=1}^r \frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} (u_j^* b) u_j. \quad (19.2)$$

The fitted-output factor has a particularly clear reduction. Put $\beta = v(\alpha)$ and $\gamma = v(\sigma_j)$, with positive leading coefficients $b_0 = \text{lc}(\alpha)$ and $a_0 = \text{lc}(\sigma_j)$. Positivity of the denominator gives

$$\text{st} \left(\frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} \right) = \begin{cases} 1, & \gamma < \beta, \\ \frac{a_0^2}{a_0^2 + b_0^2}, & \gamma = \beta, \\ 0, & \gamma > \beta. \end{cases} \quad (19.3)$$

The quotient always lies in $[0, 1]$, so its standard part exists. This is a scale filter: larger singular directions pass, smaller ones disappear in the ordinary shadow, and the boundary scale retains a nontrivial leading amplitude. At that boundary, a valuation-only description is insufficient.

Proposition 19.1 (Ordinary data and infinitesimal regularization). *If A and b have ordinary complex entries and α is a positive infinitesimal in a Hahn workspace, then x_α is finite and*

$$\text{st}(x_\alpha) = A^\dagger b,$$

where the right side is the ordinary complex pseudoinverse solution.

Proof. Choose an ordinary complex SVD. Every nonzero singular value is a positive ordinary real number. In (19.2), the coefficient $\sigma_j/(\sigma_j^2 + \alpha^2)$ is finite and has standard part $1/\sigma_j$. There are finitely many terms, so reduction gives the pseudoinverse formula. $\square$

The ordinary-data hypothesis matters. For $A = (t)$, $b = 1$, and $\alpha = t^2$, the regularized solution is $t^{-1}/(1 + t^2)$, which is infinite and has no standard part. Nor does a universal choice of one infinitesimal α resolve every surreal scale: Example 12.3 already exhibits a direction smaller than all finite powers of a chosen basic infinitesimal.

19.2 Schmidt decomposition and infinitesimal components

Give the finite tensor space $K^m \otimes_K K^n$ the inner product for which the coordinate tensor basis is orthonormal. A vector

$$\psi = \sum_{a,b} A_{ab} e_a \otimes f_b$$

is identified with its coefficient matrix A . The SVD immediately gives

$$\psi = \sum_{j=1}^r \sigma_j u_j \otimes \bar{v}_j, \quad \|\psi\|^2 = \sum_{j=1}^r \sigma_j^2. \quad (19.4)$$

Conjugation on the second vector is required by the convention $A = U\Sigma V^*$. The least number of simple tensors in a representation of ψ is its matrix rank: a sum of k simple tensors gives a matrix of rank at most k , and (19.4) attains $k = r$.

For a unit vector, the eigenvalues of the positive trace-one matrix AA^* are the σ_j^2 and zeros. Thus the finite algebraic parts of Schmidt rank and reduced-matrix spectral calculations make sense over K . For example,

$$\psi_t = \frac{e_0 \otimes f_0 + t e_1 \otimes f_1}{\sqrt{1 + t^2}}$$

has exact tensor rank two, whereas its standard part is the rank-one tensor $e_0 \otimes f_0$. Its second squared Schmidt coefficient is the positive infinitesimal $t^2/(1 + t^2)$, not zero. This illustrates precisely what is lost on taking an ordinary shadow.

This is a finite algebraic application, not a proposed physical interpretation of surreal probabilities. No infinite tensor product, measure-valued probability theory, entropy extension, or quantum dynamics is supplied by these calculations.

20 What the chapter establishes, and what remains separate

The finite-dimensional order theory is robust: orthogonality, spectral and singular-value decompositions, least squares, low-rank optimization, and the perturbation bounds proved above require real closedness, not Archimedean completeness. Their proofs exhibit the relevant witnesses, so questions about class-sized compactness never arise.

The scale theory contains additional information not visible in an ordinary complex matrix. Nonzero singular values have a finite ordered list of valuations. Determinantal minima recover that list, positive Gram coefficients recover its leading amplitudes in blocks, and the input filtration gives an intrinsic meaning to rank at a chosen resolution. The same data govern conditioning and perturbation certificates. The minimum-valuation elimination gives a finite algebraic route to those invariants when the scalar representation supplies the required exact operations.

Divisibility of the value group turns out to be separable from all of this. The decompositions survive its removal, with every exponent confined to the subgroup generated by the input

exponents and with a finite splitting tree of at most $n - 1$ nontrivial nodes, a bound independent of the order type of the supports and of the rank or cardinality of Γ . What does not survive its removal is the free availability of positive roots, and the determinantal invariant measures the deficit exactly: the field of definition of $A^{1/q}$, its degree, the criterion for no extension at all, the matching criterion for a Gram factorization, and a single scalar generator of the whole extension. That the same Δ_k answers both the scale question and the arithmetic one is the point at which the two halves of this article meet.

Several nearby subjects remain distinct. There is no theorem here about infinite-dimensional Hilbert spaces over $\mathbf{No}[i]$, bounded-operator spectra, compact or trace-class operators, or spectral measures. A support-controlled Hahn expansion of an eigenbasis *is* now available for a single matrix, over an arbitrary set-sized value group ([Theorems 9.12](#) and [9.18](#) and [Corollary 9.15](#)). There is still no global analytic choice of eigenvectors through collisions or across multiple parameters, and [Warning 9.24](#) shows that this is not a gap in the argument but a genuine failure in the multivariable formal power-series category: the crossing matrix [\(9.13\)](#) has no eigenvalue in $\mathbb{C}[[x, y]]$ at all. The descent theorems resolve the totally ordered scale problem and not that one. Nor do the exact decompositions automatically provide effective algorithms for arbitrary surreal inputs; [Section 13.3](#) lists what a representation must supply, and [Warning 13.2](#) records that even the ramification criterion can outrun a black-box order interface. Finally, the strong matrix series of [Theorem 18.1](#) is not a claim of fine-topological convergence, and neither is the splitting tree: [Example 9.2](#) exhibits a value group in which a valid Hahn identity has non-convergent partial sums, so a remainder bound is never used here as a convergence proof.

All field and extension statements here are set-sized. No proper-class Hahn sum, no proper-class vector space, and no class-sized workspace occurs anywhere; [Proposition 2.3](#) is what keeps the surcomplex statements inside a set.

Three further questions are concrete enough to name. One is to replace the finite tree of set-valued supports by finite effective support grammars for useful transseries subclasses. Another is to characterize smaller non-Hahn subfields closed under the separated-block construction, so that the full Hahn field can be replaced by a more economical domain. A third is to ask which other positive spectral functionals, beyond rational powers and their strictly positive sums, furnish similarly exact primitive generators. These are questions suggested by the results, not advertised as previously published open problems.

A formalization would have a natural order, which is worth recording because it is *not* a report of code that has been compiled: positive sum-of-squares valuation ([Lemma 9.8](#)) and integral unitary invariance ([Lemma 7.3](#)) first; then finite-index Hahn coset decompositions, the monomial character action, and the positive-coset primitive-element argument ([Lemmas 10.2](#) and [11.1](#)), all of which are set-sized and independent of the full surreal construction; then an interface for finite-variable formal series evaluated at Hahn infinitesimals, with the universal separated-block projectors and their polar intertwiner, where the ordinary complex contour justification in [Theorem 9.9](#) could be replaced by formal resolvent identities or a formal Hensel block-lifting theorem; then dimension induction for the splitting tree; and finally the normal-form embeddings. Nothing in this article has been proof-assistant verified.

For the repository, this report sits at `docs/surcomplex/spectral-theory/`. It follows the workspace conventions of the foundations and polynomial-algebra reports and can be read independently of the analytic coefficient-ring machinery. The most useful cross-reference from polynomial algebra is the bridge from its Newton profiles to [Theorems 8.1](#) and [8.2](#); the most useful cross-references to computer algebra are the exact-operation contract in [Section 13](#) and its two extensions in [Section 13.3](#), where [Warning 13.2](#) names a decision that needs a capability tier the black-box order interface does not provide [\[4\]](#).

A A proof and hypothesis ledger

Result	Scalar hypotheses	Essential finite mechanism
Spectral theorem and SVD	Real closed F , $K = F(i)$	Algebraic roots, positivity, finite orthogonal complements
Variational and low-rank extrema	Same	Eigenbasis witnesses and dimension intersection, not compactness
Least squares and pseudoinverse	Same	Orthogonal coordinates and finite sums of nonnegative squares
Singular scales from minors	Divisible Hahn workspace	Integral unitary factors and Cauchy–Binet
Leading singular amplitudes	Same, real/complex residue field	Positive sum of squared minor coefficients
Rank filtration	Same, chosen integral lattice	Lattice preservation by unitaries and coordinate reduction
Scale elimination	Exact field and valuation operations	Minimum pivot, integral operations, Schur complement
Inverse perturbation	Real closed F ; $q < 1$	A positive lower singular bound and a finite inverse identity
Visible-rank stability	Hahn workspace; strict precision threshold	Ordered singular-value Lipschitz bound plus leading-term preservation
Spectral-subspace bound	Real closed F ; separated selected/complementary spectra	Entrywise Sylvester equation and Frobenius norm
Infinite matrix series	Hahn workspace; $v(E) > 0$	Positive support lemma, not cofinality of integer multiples
Positive norm square without real closure	Real coefficients, leading order, positive-support binomial evaluation	Square roots of <i>all</i> positive field elements are not assumed
Separated-block lift	Distinct ordinary residue eigenvalues, finite-variable formal identities, positive Hahn support	Not sequential valuation convergence, not a surreal contour integral
Hermitian spectral descent	Exact trace subtraction, ordinary constant spectral theorem, block lift, induction on dimension	Divisibility of Γ , real closedness of F_Γ , finite rank
Normal and simultaneous descent	Hermitian descent, commutant argument, finite-dimensional refinement	A pre-existing algebraic closure of K_Γ , or an infinite product of unitaries
SVD without divisibility	Hermitian Gram diagonalization, norm-square roots, finite Gram–Schmidt	Square roots of arbitrary positive eigenvalues of arbitrary positive matrices

Result	Scalar hypotheses	Essential finite mechanism
Exact root field	Scalar binomial roots, spectral descent, finite-index coset basis	A generic-eigenvalue assumption, nonzero determinant, or simple spectrum
Primitive trace	Roots of unity, coset character action, <i>strictly positive</i> coefficients	Generic weights, independent eigenvalue classes, or an arbitrary square-root branch
Surcomplex interpretation	Set-supported normal forms, set-sized input subgroup	Proper-class sums or vector spaces

The order-theoretic arguments also apply directly to real symmetric matrices over F . They do not apply to an arbitrary algebraically closed field equipped with an arbitrary involution: positivity of x^*x is the indispensable additional structure. Conversely, the valuation-ring elimination argument itself requires no conjugation or positivity; those structures are what identify its invariant factors with singular values. This separation helps avoid attributing a positivity theorem to general valued-field linear algebra.

B Reproducible finite checks

The archive contains `code/verify_examples.py`, an exact SymPy verification program, together with its recorded JSON and text reports. The valuation and elimination kernel uses rational functions over $\mathbb{Q}(i)$ in two formal positive monomials t, u , with the lexicographic valuation convention

$$v(t) = (0, 1), \quad v(u) = (1, 0).$$

These correspond to the exponents 1 and ω in the subgroup $\mathbb{Z} + \mathbb{Z}\omega$. A rational function is first canceled exactly; the valuation is the least lexicographic monomial exponent of its numerator minus that of its denominator. This finite rational-function field embeds in a divisible Hahn workspace, but the code is not a general Hahn-series engine. Separate finite identity checks also use exact square-root expressions for the two-by-two spectral examples.

The checks compare minimum-valuation elimination with exhaustive minors on small exact matrices, including matrices obtained by integral invertible row and column operations from prescribed diagonal scales. They also check the positive Gram coefficient identity, the residual polynomial examples, Penrose identities for a rank-deficient matrix, the exact projector and rotation examples, the second-order spectral expansion, the nilpotent pseudospectral example, and the regularization and tensor identities. All symbolic assertions must pass for the program to write a successful report; no floating-point sampling is used.

The bundled run passed 176 exact assertions across 18 matrix-profile cases (Python 3.13.5, SymPy 1.14.0). The recorded run is a reproducibility check of these finite identities and of the implementation on its stated test cases. It is not verification of the general mathematical proofs, of arbitrary ordered groups or Hahn supports, or of proper-class foundations. In particular, a finite test cannot prove the support lemma or certify a general surreal equality algorithm. The exact assertion count and software versions are recorded in `data/verification_report.json` rather than being used as a proxy for mathematical correctness.

The second check suite

The material of [Sections 9 to 11](#) arrived with its own exact SymPy program, also with no floating-point sampling. It exercises the resolvent-projector construction of (9.4) and its polar

intertwiner (9.5) through formal degree six for a three-dimensional Hermitian example with a repeated leading eigenvalue; it tests the displayed scalar and two-by-two identities; it checks minor-valued profiles under rational unitary changes of basis, including rank-deficient profiles; and it tests finite quotient computations, subgroup enumeration against a maximal-minor lattice index, and character and stabilizer counts in finite integer lattices, the last of these being the computational face of (10.11) and (11.3).

The delivered run passed **807 exact scalar assertions** — 378 projector and polar-series coefficients, 144 finite ramification-group checks, 105 rational-matrix checks, 96 character checks, 55 block and low-degree terms, 29 displayed identities — across 48 finite quotient cases and six rational positive semidefinite profiles, under Python 3.13.5 and SymPy 1.14.0. It was rerun for this collection with an explicit output directory and reproduced PASS with exactly 807 assertions and the identical category split under Python 3.14.4 and SymPy 1.14.0.

The same caveats apply, and one more. The counts include individual zero coefficients, so they are not counts of independent theorems. The checks are finite identities and examples only: there is no exhaustive computation over arbitrary Hahn supports or arbitrary ordered groups, and the general support and field-theoretic results rest on the proofs in this article and not on the scripts. Both programs default their output directory to the shipped `data/`, so both must be run on a copy or with an explicit output path; the repository README records the safe invocations.

C Notation at a glance

Symbol	Meaning
$F, K = F(i)$	A real closed ordered field and its algebraic complexification
F_Γ, K_Γ	Real and complex Hahn workspaces, $t^\gamma = \omega^{-\gamma}$
A^*	Conjugate transpose
$\sigma_1 \geq \cdots \geq \sigma_r > 0$	Nonzero singular values, in decreasing ordered magnitude
$\ A\ _{\text{op}}, \ A\ _F$	Operator and Frobenius norms, valued in $F_{\geq 0}$
$v(a), \text{lc}(a)$	Least Hahn exponent and coefficient at that exponent
$v(A)$	Minimum valuation of the entries; $v(0) = \infty$
γ_j	$v(\sigma_j)$, in nondecreasing order
Δ_k	Minimum valuation of a k -minor, with $\Delta_0 = 0$
$\mathcal{O}, \mathfrak{m}, \text{st}$	Integral valuation ring, positive-valuation ideal, residue map
c_k	Coefficient of s^k in $\det(I + sA^*A)$
C_k	Positive leading coefficient of c_k , after monomial normalization
$r_{<\theta}, r_{\leq\theta}$	Numbers of nonzero singular scales below or at a cutoff
$A^\dagger$	Moore–Penrose inverse
$\kappa(A)$	Ordered spectral condition number of an invertible matrix
$O_v(t^\eta)$	A formal remainder with valuation at least η , not a real limit
$\Gamma_{\mathbb{Q}}$	The divisible hull $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, with its unique compatible order
$\langle S \rangle_{\mathbb{N}}$	The additive monoid generated by a well-ordered $S \subseteq \Gamma_{>0}$
H	The exact input support subgroup of a matrix, not assumed divisible
$\Gamma_{A,q}$	The root group $\Gamma + \sum_k \mathbb{Z}\Delta_k(A)/q$
$R_q(A)$	The subgroup of $\Gamma/q\Gamma$ generated by the classes $\Delta_k(A) + q\Gamma$
μ_q	The group of q th roots of unity in $\mathbb{C}^\times$
$K(B)$	The field obtained from K by adjoining every entry of a finite matrix B
$O_{\text{formal}}(E^k)$	A support-certified sum of monomials of formal degree at least k

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